

On the prime number theorem to results in gaps between consecutive primes

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1 Introduction

Some key notations include $\Re$ and $\Im$ to denote the *real* and *imaginary* parts of a complex number. We define $\pi(x)$ to be the *prime counting function* which outputs the count of prime numbers less than or equal to x . p_n denotes the n^{th} prime number. $\zeta(z)$ refers to the Riemann zeta function, as defined in Definition 3.1. $\psi(x)$ represents the second Chebyshev function, as defined in Definition 3.13.

The aim of this essay is to formally state and prove the Prime Number Theorem (PNT) by means of both analytic and elementary arguments. Subsequently, we aim to provide insight into the implications surrounding the quantity $p_{n+1} - p_n$. Informally, the PNT states that $\pi(x)$ can be approximated by $\frac{x}{\log x}$ as $x \rightarrow \infty$. Notationally, this can be seen as

$$\pi(x) \sim \frac{x}{\log x} \iff \lim_{x \rightarrow \infty} \frac{\pi(x) \log(x)}{x} = 1.$$

The PNT was originally proven independently by la Vallee Poussin and Hadamard in 1896. Their proofs relied on showing that $\zeta(z)$ has no zeros on the line $\Re(z) = 1$. Our first proof will constitute of precisely this traditional method. After becoming familiar with all the components of the traditional proof, we will proceed to outline an elementary proof attributed to Selberg and Erdős. However, before doing so we embark with a result from Chebychev in 1852 (stated from LeVeque [14, p. 148]) which states there exists positive finite constants c_1 and c_2 such that for $x \geq 2$,

$$c_1 \frac{x}{\log x} \leq \pi(x) \leq c_2 \frac{x}{\log x}.$$

This result can be regarded as a weaker version of the PNT, as it demonstrates the existence of such constants, whereas the PNT asserts that c_1 and c_2 can approach 1 arbitrarily closely. The PNT has numerous consequences for results concerning the gaps between primes and for determining the order of magnitude of other number theoretic objects, as will be demonstrated. In proving the PNT, we utilise the Riemann zeta function and its profound properties, alongside contours from complex analysis, to unveil asymptotic relationships between various analytic objects, including the zeta function, and fundamental number theoretic functions such as the prime counting function $\pi(x)$ and the second-kind Chebychev function $\psi(x)$.

One of the motivations for proving the Prime Number Theorem is its useful equivalent forms, with one of particular interest being $p_n \sim n \log n$. This consequently leads to a conditional bound on the quantity $g_n := p_{n+1} - p_n$. More precisely, we will show for any $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ we have $g_n < \epsilon \cdot p_n$. Building upon this, we will examine enhanced results regarding conditional bounds on $p_{n+1} - p_n$, and explore the implications of assuming certain conjectures such as Cramér's conjecture and the Lindelöf hypothesis. Through computational analysis, we conjecture that $g_n < p_n^{0.7125} + \frac{1}{4}$ for all $n \in \mathbb{N}$, and establish connections with existing stronger conjectures such as Andrica's conjecture. During our journey, we also encounter an intriguing result demonstrating the existence of primes between consecutive cubes for sufficiently large n . While it remains challenging to ascertain the behavior of g_n for large values of n , it becomes apparent that under the assumption of Cramér's conjecture, the expression $\sqrt{p_{n+1}} - \sqrt{p_n} \rightarrow 0$ as $n \rightarrow \infty$.

2 Chebychev's *Almost* Prime Number Theorem

Before delving into the proofs surrounding the celebrated Prime Number Theorem, let us begin with some history. Adrien-Marie Legendre (1752-1833) is credited with originally conjecturing the PNT. Here, we offer a translation of his observation, as extracted from Klyve [12, p.1]:

On a very remarkable law observed in the enumeration of prime numbers

Although the sequence of prime numbers is extremely irregular, one can however find, with a very satisfying precision, how many of these numbers there are from 1 up to a given limit x . The formula that resolves this question is

$$y = \frac{x}{\log x - 1.08366} [\dots]$$

— LEGENDRE, 1808

Pafnuty Chebychev (1821-1894) was aware of Legendre's conjecture and went further to speculate that the quotient approached arbitrarily close to 1 for large values of x . However, Chebychev was unable to prove this. Instead he proved a weaker result showing the order of magnitude of $\pi(x)$ is $\frac{x}{\log x}$.

Theorem 2.1 (Chebychev, 1852). *There exists positive finite constants c_1 and c_2 such that for sufficiently large x ,*

$$c_1 \frac{x}{\log x} \leq \pi(x) \leq c_2 \frac{x}{\log x}.$$

Moreover, Chebychev was able to show $c_1 > 0.92\dots$ and $c_2 < 1.105\dots$. These were the first theorems about $\pi(x)$ since Euclid's time. (LeVeque [14, p. 149]).

We prove Theorem 2.1 for weaker constants $c_1 = \frac{\log 2}{2} \approx 0.346$ and $c_2 = 6 \log 2 \approx 4.158$, which still satisfy Chebychev's theorem for $x \geq 2$. This approach emphasises the methods employed by Chebychev while avoiding technical details that only serve to tighten the bound. Let us first prove the upper bound.

Proof of upper bound. Chebychev's approach was to seek a combination of factorials in which the exponents on the prime power factors would be small, allowing the combination to closely approximate the product of all primes in an interval. We make use of a simpler combination than that of Chebychev's, namely

$$N := \binom{2n}{n}.$$

This proof is adapted from Lemmermeyer [13, p. 78] with certain modifications and additional justification stated throughout. Lemmermeyer begins by showing three facts about N . We realise only two of these are required for the proof. Firstly, we have the following inequality

$$\frac{2^{2n}}{2n} \leq \binom{2n}{n} \leq 2^{2n}. \quad (1)$$

Both inequalities follows from the Binomial Theorem. Certainly, we have $\binom{2n}{n}$ is one term of the sum $\sum_{m=0}^{2n} \binom{2n}{m} = (1+1)^{2n} = 2^{2n}$ so indeed the right-hand side of the inequality in (1)

holds. For the left-hand side, we have

$$2^{2n} = (1+1)^{2n} = \sum_{m=0}^{2n} \binom{2n}{m} = 2 + \sum_{m=1}^{2n-1} \binom{2n}{m} \leq 2n \cdot \binom{2n}{n},$$

where the final inequality makes use of $\binom{2n}{n}$ being the largest term in the sum. Dividing through by $2n$ yields the left-hand side.

Secondly, by the expanding the binomial coefficient, one can observe that,

$$\binom{2n}{n} \text{ is divisible by all primes } n < p \leq 2n. \quad (2)$$

We can then apply (2) and (1) respectively to get

$$n^{\pi(2n)-\pi(n)} \leq \prod_{n < p \leq 2n} p \leq \binom{2n}{n} \leq 2^{2n}.$$

Taking logs gives

$$\pi(2n) - \pi(n) \leq \log_n(2^{2n}) = \frac{\log(2^{2n})}{\log(n)} = 2(\log 2) \frac{n}{\log n}. \quad (3)$$

At this point, we see an upper bound has emerged. We claim that

$$\pi(2^k) \leq \frac{3 \cdot 2^k}{k}, \quad (4)$$

for all $k \geq 1$. The cases for $k \leq 5$ can be verified separately. Suppose $k > 5$, we then induct on k . Assume $\pi(2^k) \leq \frac{3 \cdot 2^k}{k}$, then we have

$$\pi(2^{k+1}) \leq \pi(2^k) + 2 \log(2) \frac{2^k}{\log 2^k} = \pi(2^k) + \frac{2 \cdot 2^k}{k} \leq \frac{3 \cdot 2^k}{k} + \frac{2 \cdot 2^k}{k} = \frac{5 \cdot 2^k}{k} \leq \frac{3 \cdot 2^{k+1}}{k+1},$$

where the first inequality is by applying (3) with $n = 2^k$, the second by our inductive hypothesis and the final by our assumption of $k > 5$. This proves the claim.

Now suppose $4 \leq 2^k < x \leq 2^{k+1}$, then using (4), we have

$$\pi(x) \leq \pi(2^{k+1}) \leq \frac{6 \cdot 2^k}{k+1} \leq 6 \log 2 \cdot \frac{2^k}{\log 2^k} \leq 6 \log 2 \frac{x}{\log x},$$

where the penultimate inequality comes from the following set of implications: $\log 2^k \leq \log 2^k + \log 2 \implies \log 2^k \leq (\log 2)(k+1) \implies \frac{1}{k+1} \leq \frac{\log 2}{\log 2^k}$, and the final inequality is because $\frac{x}{\log x}$ is a monotonically increasing function. We can also show by computation that $\pi(x) \leq 6 \log 2 \frac{x}{\log x}$ for $x < 4$. These series of justifications is additional to what Lemmermeyer provides.

This completes the upper bound. $\square$

For the proof of the lower bound, we are required to introduce the notion of *p-adic valuation*.

Definition 2.2. Given a prime p and an integer n , let $A = \{m \in \mathbb{N} : p^m \mid n\}$. Then the *p-adic valuation* of n at p is defined as $v_p(n) = \max A$.

Remark 2.3. We will assume without proof basic properties of $v_p(n)$ which are

1. $v_p\left(\frac{r}{s}\right) = v_p(r) - v_p(s)$ this can be viewed as an extension into $\mathbb{Q}$ with $s \neq 0$,
2. $v_p(nm) = v_p(n) + v_p(m)$.

We state a useful lemma which we will use for our proof of the lower bound. This is known as *Legendre's Formula*.

Lemma 2.4 (Legendre's Formula). *With $v_p(n)$ defined in 2.2, we have*

$$v_p(n!) = \sum_{m \geq 1} \left\lfloor \frac{n}{p^m} \right\rfloor.$$

The lemma can be easily proved by inducting on n , we omit the proof for our purposes. We now continue with our proof of Theorem 2.1.

Proof of lower bound. Applying Legendre's formula to $N = \binom{2n}{n}$ and using the properties of $v_p(n)$ given in Remark 2.3 we have

$$v_p(N) = v_p((2n)!) - (v_p(n!) + v_p(n!)) = \sum_{m \geq 1} \left\lfloor \frac{2n}{p^m} \right\rfloor - \left(2 \sum_{m \geq 1} \left\lfloor \frac{n}{p^m} \right\rfloor \right) = \sum_{m \geq 1} \left(\left\lfloor \frac{2n}{p^m} \right\rfloor - 2 \left\lfloor \frac{n}{p^m} \right\rfloor \right). \quad (5)$$

Then for $m > \frac{\log(2n)}{\log(p)} \iff \log(p^m) > \log(2n) \iff p^m > 2n$, we have $\left\lfloor \frac{2n}{p^m} \right\rfloor = 0$ and since $p^m > 2n > n$ we also have $\left\lfloor \frac{n}{p^m} \right\rfloor = 0$. Therefore,

$$\left\lfloor \frac{2n}{p^m} \right\rfloor - 2 \left\lfloor \frac{n}{p^m} \right\rfloor = 0. \quad (6)$$

A property of the floor function is, for all $x \in \mathbb{R}$ we have $\lfloor 2x \rfloor - 2\lfloor x \rfloor \in \{0, 1\}$. Applying this property alongside (6) to our result in (5), we have

$$v_p(N) = \sum_{m=1}^{\left\lfloor \frac{\log(2n)}{\log(p)} \right\rfloor} \left(\left\lfloor \frac{2n}{p^m} \right\rfloor - 2 \left\lfloor \frac{n}{p^m} \right\rfloor \right) \leq \sum_{m=1}^{\left\lfloor \frac{\log(2n)}{\log(p)} \right\rfloor} 1 = \left\lfloor \frac{\log(2n)}{\log(p)} \right\rfloor. \quad (7)$$

Whilst Lemmermeyer makes the jump from (5) to (7), we provide the intermediate steps to justify this. From (1), we have the following inequality

$$2n \log(2) - \log(2n) = \log\left(\frac{2^{2n}}{2n}\right) \leq \log\left(\frac{2n}{n}\right).$$

Then, by applying the fundamental theorem of arithmetic to $N = \binom{2n}{n}$ and making use of the additive property for logarithms, we have

$$\log \binom{2n}{n} = \log \prod_{p \leq 2n} p^{v_p(N)} = \sum_{p \leq 2n} \log(p^{v_p(N)}).$$

Therefore, using the bound in (7) and a property of the floor function namely $\lfloor x \rfloor \leq x$, we have

$$\sum_{p \leq 2n} \log(p^{v_p(N)}) = \sum_{p \leq 2n} v_p(N) \log(p) \leq \sum_{p \leq 2n} \left\lfloor \frac{\log(2n)}{\log(p)} \right\rfloor \log(p) \leq \sum_{p \leq 2n} \log(2n) = \pi(2n) \cdot \log(2n),$$

where the final equality follows from the definition of $\pi(x)$. Thus we have shown $2n \log 2 - \log 2n \leq \pi(2n) \cdot \log(2n)$, from which we deduce

$$\pi(2n) \geq \log(2) \cdot \frac{2n}{\log(2n)} - 1. \quad (8)$$

We finally claim that (8) implies that $\pi(x) \geq \frac{\log 2}{2} \frac{x}{\log x}$ for all $x \geq 2$. One can verify the result for $2 \leq x \leq 16$. For $x > 16$, pick $n \in \mathbb{Z}$ such that $16 \leq 2n < x \leq 2n + 2$. Then, we first observe

$$\frac{2n}{\log 2n} - \frac{n+1}{\log 2n} = \frac{n-1}{\log 2n} \geq \frac{7}{4 \log 2} \geq \frac{1}{\log 2}. \quad (9)$$

Hence, by applying both (8) and (9), we have

$$\begin{aligned} \pi(x) &\geq \pi(2n) \geq \log(2) \cdot \frac{2n}{\log(2n)} - 1 \geq \log(2) \cdot \left(\frac{1}{\log 2} + \frac{n+1}{\log 2n} \right) - 1 = \log(2) \cdot \frac{n+1}{\log 2n} \\ &\geq \log(2) \cdot \frac{n+1}{\log(2n+2)} \\ &= \frac{\log(2)}{2} \frac{2(n+1)}{\log(2(n+1))} \\ &\geq \frac{\log(2)}{2} \frac{x}{\log x}, \end{aligned}$$

where the final inequality again makes use of $\frac{x}{\log x}$ being a monotonically increasing function and $2n+2 \geq x$. Thus we have proved the lower bound. $\square$

As mentioned in the introduction, Chebychev's theorem can be considered as a weaker version of the PNT since it only asserts the existence of c_1 and c_2 . To tighten the bounds requires more technical proofs. Most notably, in 1892 Sylvester [18, C.72] was able to show

$$0.956 \frac{x}{\log x} < \pi(x) < 1.045 \frac{x}{\log x},$$

for sufficiently large x . The PNT however tells us that c_1 and c_2 can be taken arbitrarily close to 1 thus giving us the asymptotic relationship. The subsequent sections lay the groundwork to prove the PNT, which serves as a powerful tool for establishing numerous results concerning gaps in consecutive prime numbers.

3 Theory of the Prime Number Theorem

We commence with the words of Diamond [5, p.3], who reflects on the attempts made up until the time of Chebychev in 1852 to prove the PNT.

The approach of Sylvester was ad hoc and computationally complex; it offered no hope of leading to a proof of the P.N.T. Indeed, Sylvester concluded in his article with the lament that “...we shall probably have to wait [for a proof of the P.N.T.] until someone is born into the world so far surpassing Tchebychef in insight and penetration as Tchebychef has proved himself superior in these qualities to the ordinary run of mankind.

— HAROLD G. DIAMOND, 1982

Only four years after Sylvester, the PNT was proven. The traditional analytical proof of the prime number theorem heavily relies on complex analysis. Therefore, we will closely follow the approach outlined by Stein and Shakarchi [17, c. 7], occasionally incorporating proofs from external sources while providing our own reasoning for doing so.

3.1 The Riemann Zeta function

The goal of this subsection is to prove two useful results: Corollary 3.5, which allows us to derive a contradiction in the proof of showing ζ has no zeros on $\Re(z) = 1$; Lemma 3.6, which derives an expression for $\frac{\zeta'}{\zeta}$ for proving a key ingredient used in the proof of the PNT.

While we make extensive use of Stein and Shakarchi [17], we significantly rearrange the order in which the results are obtained to better illustrate why each result is important and how it becomes useful later on.

Definition 3.1. The *Riemann zeta function* is defined as

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}, \quad \text{where } \Re(z) > 1.$$

In subsequent proofs, the product form of the zeta function proves to be more advantageous.

Lemma 3.2 (Euler product form). *For $\Re(z) > 1$, the Riemann zeta function can be equivalently expressed as*

$$\zeta(z) = \prod_p \frac{1}{1 - p^{-z}},$$

where the product is taken over all primes p in $\mathbb{N}$.

The proof presented by Harper [8, p. 3] offers a more direct approach compared to that in Stein and Shakarchi [17, c. 7]. In the latter, the fundamental theorem of arithmetic is employed twice in a manner that obscures the construction of this formula from definition.

Proof. Let $z \in \mathbb{C}$ such that $\Re(z) > 1$. By the geometric series, we have

$$\sum_{k=0}^{\infty} \frac{1}{p^{kz}} = \sum_{k=0}^{\infty} \left(\frac{1}{p^z} \right)^k = \frac{1}{1 - p^{-z}}, \quad \text{for any prime } p.$$

The above is absolutely convergent since $\Re(z) > 1$. We now observe the following, let P be a fixed prime number. Then,

$$\prod_{p \leq P} \frac{1}{1 - p^{-z}} = \prod_{p \leq P} \sum_{k=0}^{\infty} \frac{1}{p^{kz}} = \sum_{n=1}^{\infty} \frac{c_P(n)}{n^z}, \quad \text{where } c_P(n) = \begin{cases} 1, & \text{if all prime factors of } n \leq P, \\ 0, & \text{otherwise.} \end{cases}$$

The last equality comes from the absolute convergence of the summation, which lets us rearrange terms in a finite product of an absolutely convergent series, and an application of the fundamental theorem of arithmetic. Hence,

$$\left| \zeta(z) - \prod_{p \leq P} \frac{1}{1 - p^{-z}} \right| = \left| \sum_{n=1}^{\infty} \frac{1}{n^z} - \sum_{n=1}^{\infty} \frac{c_P(n)}{n^z} \right| = \left| \sum_{\substack{n=1, \\ c_P(n)=0}}^{\infty} \frac{1}{n^z} \right| = \sum_{\substack{n=1, \\ c_P(n)=0}}^{\infty} \frac{1}{n^{\Re(z)}}.$$

Since $c_P(n) = 1$ for $n \leq P$, it follows

$$\sum_{\substack{n=1, \\ c_P(n)=0}}^{\infty} \frac{1}{n^{\Re(z)}} \leq \sum_{\substack{n=P+1, \\ c_P(n)=0}}^{\infty} \frac{1}{n^{\Re(z)}} \xrightarrow[\text{if } \Re(z) > 1]{\text{as } P \rightarrow \infty} 0.$$

Therefore, $\zeta(z) = \prod_p \frac{1}{1 - p^{-z}}$, as required. $\square$

This result immediately creates a connection between the Riemann zeta function and primes, hinting at the potential utility of the zeta function in proving results pertaining to primes such as the PNT. We may now analyse this variation of the zeta function.

The following result, Stein and Shakarchi [17, C7. Lemma 1.3], is obtained by applying the Taylor expansion to the logarithm function and using the Euler product form of the zeta function thereby allowing us to use the product rule for logarithms.

Lemma 3.3 (Logarithm of zeta function). *Suppose $z \in \mathbb{C}$ such that $\Re(z) > 1$, then*

$$\log \zeta(z) = \sum_p \sum_{m=1}^{\infty} \frac{p^{-mz}}{m} = \sum_{n=1}^{\infty} c_n n^{-z},$$

for some $c_n \geq 0$.

Proof. We use the following Taylor expansion,

$$\log \left(\frac{1}{1 - z} \right) = \sum_{m=1}^{\infty} \frac{z^m}{m} \quad \text{for } 0 \leq \Re(z) < 1.$$

Using the Euler product variant of the zeta function from Lemma 3.2, and the multiplicative property of logarithms, we have

$$\log \zeta(z) = \log \prod_p \frac{1}{1 - p^{-z}} = \sum_p \log \left(\frac{1}{1 - p^{-z}} \right) = \sum_p \sum_{m=1}^{\infty} \frac{p^{-mz}}{m}, \quad (10)$$

where we apply the Taylor expansion to the last equality. Since $\zeta(z)$ is holomorphic for all $z \in \mathbb{C}$ except at the simple pole $z = 1$, and has no zeros for $\Re(z) > 1$, one can analytically extend (63) to hold for all $z \in \mathbb{C}$ such that $\Re(z) > 1$, by applying Stein and Shakarchi [17, C3. Theorem 6.2]. We can reformulate the above as $\sum_{n=1}^{\infty} c_n n^{-z}$, where $c_n = \frac{1}{m}$ if $n = p^m$ and $c_n = 0$ otherwise. $\square$

We can apply this property of ζ in a crude way to get Corollary 3.5. This is useful because in Theorem 3.12 we construct a proof by contradiction to show ζ has no zeros on $\Re(z) = 1$. As discussed in the introduction, this constitutes a crucial step towards proving the PNT. The following Lemma corresponds to Stein and Shakarchi [17, C7. Corollary 1.5].

Lemma 3.4. *Let $\sigma, t \in \mathbb{R}$ such that $\sigma > 1$, then:*

$$\log |\zeta^3(\sigma) \cdot \zeta^4(\sigma + it) \cdot \zeta(\sigma + 2it)| \geq 0.$$

Proof. Let $z = \sigma + it$. Then,

$$\begin{aligned} \Re(n^{-z}) &= \Re(n^{-\sigma-it}) = \Re(e^{(-\sigma-it)\log(n)}) = \Re(e^{-\sigma\log(n)} e^{-it\log(n)}) \\ &= \Re(e^{-\sigma\log(n)} (\cos(-t\log(n)) + i \sin(-t\log(n)))) \\ &= e^{-\sigma\log(n)} \cdot \cos(t\log(n)) \\ &= n^{-\sigma} \cdot \cos(t\log(n)). \end{aligned}$$

Hence,

$$\begin{aligned} \log |\zeta^3(\sigma) \cdot \zeta^4(\sigma + it) \cdot \zeta(\sigma + 2it)| &= 3 \log |\zeta(\sigma)| + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)| \\ &= 3\Re(\log(\zeta(\sigma))) + 4\Re(\log(\zeta(\sigma + it))) + \Re(\log(\zeta(\sigma + 2it))), \end{aligned}$$

where the last equality makes use of the property $\log |z| = \Re(\log(z))$, for $z \in \mathbb{C}$. Applying Lemma 3.3 yields,

$$\begin{aligned} &= \Re \left(\sum_{n=1}^{\infty} c_n n^{-\sigma} \right) + 4\Re \left(\sum_{n=1}^{\infty} c_n n^{-(\sigma+it)} \right) + \Re \left(\sum_{n=1}^{\infty} c_n n^{-(\sigma+2it)} \right) \\ &= 3 \sum_{n=1}^{\infty} c_n n^{-\sigma} + 4 \sum_{n=1}^{\infty} c_n n^{-\sigma} \cdot \cos(t\log(n)) + \sum_{n=1}^{\infty} c_n n^{-\sigma} \cdot \cos(2t\log(n)) \\ &= \sum_{n=1}^{\infty} c_n n^{-\sigma} (3 + 4\cos(t\log(n)) + \cos(2t\log(n))) \geq 0. \end{aligned}$$

This completes the proof. $\square$

Corollary 3.5. *We have $|\zeta^3(\sigma) \cdot \zeta^4(\sigma + it) \cdot \zeta(\sigma + 2it)| \geq 1$.*

Proof. This follows directly from the above since $\log(x) < 0$ for $x \in (0, 1)$. $\square$

Also from Lemma 3.3 regarding $\log \zeta(z)$, we derive a result used later in Proposition 3.20. This result corresponds to the discussion at Stein and Shakarchi [17, pg. 191].

Lemma 3.6. Suppose $z \in \mathbb{C}$ such that $\Re(z) > 1$. Then

$$-\frac{\zeta'(z)}{\zeta(z)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^z}, \quad \text{where } \Lambda(n) := \begin{cases} \log p, & \text{if } n = p^m \text{ for some } m \in \mathbb{Z}_{\geq 0}, \\ 0, & \text{otherwise.} \end{cases}$$

$\Lambda(n)$ is known as the Von Mangoldt function.

Proof. Recall from Lemma 3.3 that $\log \zeta(z) = \sum_p \sum_{m=1}^{\infty} \frac{p^{-mz}}{m}$. Since the series is absolutely convergent, we can negate this expression and differentiate term by term to obtain our desired result. Differentiating both sides of the expression for $\log \zeta(z)$ with respect to z yields

$$-\frac{\zeta'(z)}{\zeta(z)} = \sum_p \sum_{m=1}^{\infty} -\frac{1}{m} \cdot \frac{d}{dz} \left(\frac{1}{p^{mz}} \right).$$

Therefore,

$$\sum_p \sum_{m=1}^{\infty} -\frac{1}{m} \cdot \frac{d}{dz} \left(\frac{1}{p^{mz}} \right) = \sum_p \sum_{m=1}^{\infty} -\frac{1}{m} \cdot (-m \log(p) \cdot p^{-mz}) = \sum_p \sum_{m=1}^{\infty} \frac{\log(p)}{p^{mz}}.$$

By defining $\Lambda(n)$ as above, we indeed get

$$\sum_p \sum_{m=1}^{\infty} \frac{\log(p)}{p^{mz}} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^z},$$

as required. □

3.2 Zero free region on $\Re(z) = 1$

As mentioned earlier, we utilise the theory developed thus far to derive a contradiction, establishing that ζ has no roots on $\Re(z) = 1$. To achieve this, we also require certain fundamentals from complex analysis. These proofs are motivated from MA244 Analysis III lectures.

Theorem 3.7. Suppose that f is holomorphic in a connected open set Ω , has a zero at a point $z_0 \in \Omega$ and does not vanish identically in Ω that is $f|_{\Omega} \not\equiv 0$. Then there exists a neighbourhood $U \subset \Omega$ of z_0 , a non-vanishing holomorphic function g on U and a unique positive integer n such that

$$f(z) = (z - z_0)^n g(z),$$

for all $z \in U$.

Proof. Since Ω is connected and $f \not\equiv 0$ on Ω , then $f \not\equiv 0$ in some neighbourhood of z_0 call it U . Consider a small disc, $B(z_0, r)$, centered at z_0 for some $r > 0$. Then f has a power series expansion as seen in MA244 Analysis III given by

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k.$$

Since $f \not\equiv 0$ near z_0 , there exists a smallest integer n such that $a_n \neq 0$. It then follows

$$f(z) = \sum_{k=n}^{\infty} a_k(z - z_0)^k = (z - z_0)^n [a_n + a_{n+1}(z - z_0) + \cdots] = (z - z_0)^n g(z),$$

where g is defined above by $g(z) = \sum_{i=n}^{\infty} a_i(z - z_0)^{i-n}$. We observe that g is holomorphic by construction and is non-vanishing for all z in U since $a_n \neq 0$.

For the uniqueness of n , we suppose for a contradiction that

$$f(z) = (z - z_0)^n g(z) = (z - z_0)^m h(z), \quad (11)$$

with $h(z_0) \neq 0$. If $m > n$, then we divide (11) by $(z - z_0)^n$ to obtain

$$g(z) = (z - z_0)^{m-n} h(z).$$

It follows that as $z \rightarrow z_0$, we get $g(z_0) = 0$ which is a contradiction since g is non-vanishing on U and $z_0 \in U$. For $m < n$, we can show in a symmetrical way that $h(z_0) = 0$ which is also a contradiction. Therefore $m = n$ which implies $g = h$. This establishes our result. $\square$

Motivated by the theorem above, we introduce the following definitions, as taken from Stein and Shakarchi [17, C3. p.74].

Definition 3.8. We say that f has a *zero of order n* at z_0 if we can express f as $f(z) = (z - z_0)^n g(z)$ for some non-vanishing, holomorphic function g . If a zero is of order 1, then we call it a *simple zero*.

Definition 3.9. We define the *deleted neighbourhood* of z_0 to be an open disc centered at z_0 excluding the point z_0 . This is precisely the following set

$$\{z \in \mathbb{C} : 0 < |z - z_0| < r\},$$

where $r > 0$ is the radius of the disc.

Definition 3.10. A function f defined in the deleted neighbourhood of z_0 has a *pole* at z_0 , if the function $\frac{1}{f}$, defined to be 0 at z_0 , is holomorphic in a full neighbourhood of z_0 .

This allows us to state the result in Stein and Shakarchi [17, Theorem 1.2].

Corollary 3.11. *If f has a pole at $z_0 \in \Omega$, then in a neighbourhood of z_0 , there exists a non-vanishing holomorphic function h and a unique positive integer n such that*

$$f(z) = (z - z_0)^{-n} h(z).$$

Proof. From the definition of a pole, we know that $\frac{1}{f(z)}$ is holomorphic in some neighborhood of z_0 , denoted as $U \subset \Omega$. Additionally, from the definition, we know that z_0 is a zero of $\frac{1}{f}$. Therefore, by applying Theorem 3.7, we have

$$\frac{1}{f(z)} = (z - z_0)^n g(z),$$

where g is holomorphic and non-vanishing in U . Therefore, it follows

$$f(z) = (z - z_0)^{-n} h(z),$$

where $h(z) = \frac{1}{g(z)}$. □

This brings us to the following crucial result, Stein and Shakarchi [17, Theorem 1.2], for the forthcoming proof of the PNT.

Theorem 3.12 (Zero free region on $\Re(z) = 1$). *The zeta function, $\zeta(z)$, has no zeros on the line $\Re(z) = 1$.*

Proof. Suppose for a contradiction that $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. Since ζ is holomorphic at $1 + it_0$ (see Appendix A), it must vanish to at least order 1 at this point. Therefore, by Theorem 3.7, we have

$$\lim_{\sigma \rightarrow 1} |\zeta(\sigma + it_0)|^4 \leq \lim_{\sigma \rightarrow 1} C(\sigma - 1)^4, \quad (12)$$

for some $C > 0$. We also have $z = 1$ is a simple pole for $\zeta(z)$ (also from appendix A) so from Corollary 3.11, it follows

$$\lim_{\sigma \rightarrow 1} |\zeta(\sigma)|^3 \leq \lim_{\sigma \rightarrow 1} C'(\sigma - 1)^{-3}, \quad (13)$$

for some $C' > 0$. So it follows by the product of (12) and (13)

$$0 \leq \lim_{\sigma \rightarrow 1} |\zeta^3(\sigma) \zeta^4(\sigma + it_0)| \leq CC' \lim_{\sigma \rightarrow 1} (\sigma - 1) = 0.$$

Therefore by the sandwich theorem, $\lim_{\sigma \rightarrow 1} |\zeta^3(\sigma) \zeta^4(\sigma + it_0)| = 0$. Finally, we have ζ is holomorphic at the points $\sigma + 2it_0$, so $|\zeta(\sigma + 2it_0)|$ is bounded on $\sigma \in [0, 1]$. Then we deduce

$$|\zeta^3(\sigma) \zeta^4(\sigma + it_0) \zeta(\sigma + 2it_0)| \rightarrow 0 \text{ as } \sigma \rightarrow 1.$$

This contradicts Corollary 3.5, which asserts that $|\zeta^3(\sigma) \cdot \zeta^4(\sigma + it) \cdot \zeta(\sigma + 2it)| \geq 1$. □

3.3 Chebychev's ψ function

This subsection *nearly* proves the PNT. While the PNT is an unconditional statement, we demonstrate that it must hold given the truth of a certain condition. That condition involves assuming $\psi(x) \sim x$, where $\psi(x)$ is defined below. Given this implication, we can therefore reduce the PNT to studying $\psi(x)$. The objective of this subsection is to demonstrate the validity of this implication.

Definition 3.13. Suppose $x \in \mathbb{N}$, then we define the *second Chebychev* function, $\psi(x)$, as follows

$$\psi(x) = \sum_{m=1}^{\infty} \sum_{p^m \leq x} \log p.$$

Remark 3.14. We can also express $\psi(x)$ in terms of the Von Mangoldt function (introduced in Lemma 3.6) as follows

$$\psi(x) = \sum_{m=1}^{\infty} \sum_{p^m \leq x} \log p = \sum_{m=1}^{\infty} \sum_{\substack{p, \\ p^m \leq x}} \Lambda(p^m) = \sum_{n \leq x} \Lambda(n).$$

We proceed to prove a lemma that enables us to express the second Chebychev function in a particular form, facilitating the proof of a *conditional* PNT, as presented in Theorem 3.17.

Lemma 3.15 (Equivalent definition of $\psi(x)$). *The second Chebychev function, $\psi(x)$, can be expressed as*

$$\sum_{p \leq x} \log p \left\lfloor \frac{\log x}{\log p} \right\rfloor.$$

Proof. If $x < 2$, then the result is trivial as

$$\sum_{m=1}^{\infty} \sum_{p^m \leq x} \log p = \sum_{p \leq x} \log p \left\lfloor \frac{\log x}{\log p} \right\rfloor = 0,$$

since there are no primes less than 2. Suppose $x \geq 2$. Then, if $p^m \leq x$, it follows that $m \log p \leq \log x$, and consequently $m \leq \frac{\log x}{\log p}$, for all $m \geq 1$. We utilise this alongside the ability to interchange the order of summation to get

$$\sum_{m=1}^{\infty} \sum_{p^m \leq x} \log p = \sum_{p \leq x} \sum_{\substack{m=1, \\ p^m \leq x}}^{\infty} \log p = \sum_{p \leq x} \sum_{\substack{k=1, \\ k \leq \log_p x}}^{\infty} \log p = \sum_{p \leq x} \log p \left(\sum_{\substack{k=1, \\ k \leq \log_p x}}^{\infty} 1 \right) = \sum_{p \leq x} \log p \left\lfloor \frac{\log x}{\log p} \right\rfloor,$$

where $\log_p x = \frac{\log x}{\log p}$. □

We proceed to establish the main result of this subsection, demonstrating the connection between the $\psi(x)$ function and the distribution of prime numbers. Let us first define what it means for two functions to be asymptotically equal.

Definition 3.16. The functions $f(n)$ and $g(n)$ are *asymptotically equal* if we have

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1.$$

We denote this by $f(n) \sim g(n)$.

We can now express our main theorem using this notation as seen in Stein and Shakarchi [17, C7. Proposition 2.1].

Theorem 3.17. *If $\psi(x) \sim x$, then $\pi(x) \sim \frac{x}{\log x}$.*

Proof. We show this by the following equivalence:

$$\pi(x) \sim \frac{x}{\log x} \iff \lim_{x \rightarrow \infty} \frac{\pi(x) \cdot \log x}{x} = 1 \iff \left\{ 1 \leq \liminf_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} \text{ and } \limsup_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} \leq 1 \right\}.$$

We commence by proving both inequalities in the equivalence above. First, let us employ the floor variation of the second Chebychev function as demonstrated in Lemma 3.15. Since $x \geq 1$, we have

$$\psi(x) = \sum_{p \leq x} \log p \left\lfloor \frac{\log x}{\log p} \right\rfloor \leq \sum_{p \leq x} \log p \frac{\log x}{\log p} = \sum_{p \leq x} \log x = \pi(x) \log x.$$

Then, dividing through by x , we obtain

$$\frac{\psi(x)}{x} \leq \frac{\pi(x) \log x}{x}.$$

Using $\psi(x) \sim x \iff \lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1$, we have

$$\liminf_{x \rightarrow \infty} \frac{\psi(x)}{x} \leq \liminf_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} \iff 1 \leq \liminf_{x \rightarrow \infty} \frac{\pi(x) \log x}{x},$$

which is what we require. Now, for the second inequality, fix $0 < \alpha < 1$. By leveraging Lemma 3.15 concerning the floor variation of the second Chebychev function, we observe that

$$\psi(x) = \sum_{p \leq x} \log p \left\lfloor \frac{\log x}{\log p} \right\rfloor \geq \sum_{p \leq x} \log p \geq \sum_{x^\alpha \leq p \leq x} \log p \geq \sum_{x^\alpha \leq p \leq x} \log(x^\alpha) = (\pi(x) - \pi(x^\alpha)) \log(x^\alpha).$$

Therefore, we have

$$\psi(x) + \alpha \pi(x^\alpha) \log x \geq \alpha \pi(x) \log x.$$

Since $x \geq 1$, dividing through by x , we obtain

$$\frac{\psi(x)}{x} + \frac{\alpha \pi(x^\alpha) \log x}{x} \geq \frac{\alpha \pi(x) \log x}{x}.$$

Hence,

$$\frac{\psi(x)}{x} \geq \alpha \cdot \frac{\pi(x) \log x - \pi(x^\alpha) \log x}{x}.$$

Since $\alpha < 1$, we have $\pi(x^\alpha) \leq x^\alpha$. Applying this to the above yields,

$$\frac{\psi(x)}{x} \geq \alpha \cdot \frac{\pi(x) \log x - x^\alpha \log x}{x}.$$

Then using $\psi(x) \sim x$ and $\lim_{x \rightarrow \infty} \frac{x^\alpha}{x} = 0$, we have

$$1 \geq \alpha \cdot \limsup_{x \rightarrow \infty} \frac{\pi(x) \log x}{x}.$$

Given α is arbitrary, we have the desired inequality and thus proved that $\pi(x) \sim \frac{x}{\log x}$. $\square$

The above theorem asserts that if $\psi(x) \sim x$ then the PNT holds. Therefore, we now delve into the asymptotic behavior of $\psi(x)$ as x tends to infinity.

3.4 Relating ψ to ζ

We introduce a new function that takes ψ as an argument. This will enable us to establish a clearer connection to ζ .

Definition 3.18 (Smoothing function). We define the *smoothing function*, denoted ψ_1 , by

$$\psi_1(x) = \int_1^x \psi(u) \, du.$$

We establish a relationship between ψ_1 and ζ in two steps with the first seen in Stein and Shakarchi [17, C7. Lemma 2.4].

Lemma 3.19 (Key Ingredient). *Suppose $c > 0$, then*

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{a^z}{z(z+1)} \, dz = \begin{cases} 0, & \text{if } 0 < a \leq 1, \\ 1 - \frac{1}{a}, & \text{if } a \geq 1. \end{cases}$$

It must be noted that the integral is taken over the vertical line $\Re(z) = c$.

Proof. Firstly, we have $|a^z| = a^c$. Suppose $a \geq 1$ and let $a = e^\beta$ with $\beta = \log a \geq 0$. Define

$$f(z) := \frac{a^z}{z(z+1)} = \frac{e^{\beta z}}{z(z+1)}.$$

Then it follows, $\text{Res}_{z=0} f(z) = 1$ and $\text{Res}_{z=-1} f(z) = -e^{-\beta} = -\frac{1}{a}$. Let $T > 0$ and consider the contour shown in Figure (1).

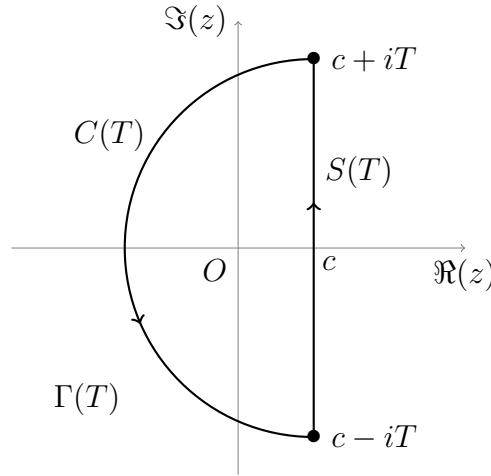


Figure 1: The contour denoted as $\Gamma(T)$.

The path $\Gamma(T)$ comprises two components: $S(T)$, a vertical line segment extending from $c - iT$ to $c + iT$, and $C(T)$, a half-circle centered at c with radius T , positioned to the left of the line segment. The contour is oriented positively (counter-clockwise). We can select T to be

sufficiently large so that 0 and -1 lie within the interior of the contour. Then, by the residue theorem

$$\frac{1}{2\pi i} \int_{\Gamma(T)} f(z) dz = \text{Res}_{z=0} f(z) + \text{Res}_{z=-1} f(z) = \left(1 - \frac{1}{a}\right).$$

We employ this variation of the residue theorem since both poles are of first order. Additionally, by construction of the contour, we have the following

$$\int_{\Gamma(T)} f(z) dz = \int_{S(T)} f(z) dz + \int_{C(T)} f(z) dz.$$

Hence, it suffices to prove that the integral over $C(T)$ approaches 0 as $T \rightarrow \infty$.

Let $s = \sigma + it \in C(T)$, then

$$|s + 1| \geq |s| \geq \frac{T}{2}.$$

Hence,

$$|s(s + 1)| \geq \frac{T^2}{4}.$$

Additionally, since $\sigma \leq c$, we have $|e^{\beta z}| \leq e^{\beta c}$. Thus, it follows

$$\left| \int_{C(T)} f(z) dz \right| \leq \frac{T a^c 4\pi}{T^2} \rightarrow 0 \text{ as } T \rightarrow \infty.$$

Therefore, we establish the result for $a \geq 1$. For $0 < a \leq 1$, a similar contour can be applied, but with the half-circle positioned to the right of the line segment. A comparable argument demonstrates that the integral over the half circle approaches 0. Moreover, as $f(z)$ has no singularities within this contour, the residue theorem asserts that the integral must be equal to 0. $\square$

The above lemma can now be extended into a second step to provide a useful connection between ψ_1 and ζ . This is precisely what Stein and Shakarchi [17, C7. Proposition 2.3] states.

Proposition 3.20 (Refined Key Ingredient). *Suppose $c > 1$, then*

$$\psi_1(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{z+1}}{z(z+1)} \left(-\frac{\zeta'(z)}{\zeta(z)} \right) dz.$$

Proof. First, observe that we have

$$\psi(u) = \sum_{n=1}^{\infty} \Lambda(n) f_n(u), \quad \text{where } f_n(u) = \begin{cases} 1, & \text{if } n \leq u, \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

The above directly follows from Remark 3.14 concerning the connection between ψ and the Von Mangoldt function, $\Lambda(n)$. Therefore, substituting (14) into the definition of $\psi_1(x)$ yields

$$\psi_1(x) = \int_0^x \psi(u) du = \int_0^x \sum_{n=1}^{\infty} \Lambda(n) f_n(u) du.$$

Using the additive property of integrals, we have

$$\int_0^x \sum_{n=1}^{\infty} \Lambda(n) f_n(u) \, du = \sum_{n=1}^{\infty} \int_0^x \Lambda(n) f_n(u) \, du.$$

Then, since $f_n(u) = 1$ for $u \geq n$, we have

$$\sum_{n=1}^{\infty} \int_0^x \Lambda(n) f_n(u) \, du = \sum_{n=1}^{\infty} \Lambda(n) \cdot \int_n^x du = \sum_{n=1}^{\infty} \Lambda(n) \cdot (x - n).$$

So we have shown that

$$\psi_1(x) = \sum_{n=1}^{\infty} \Lambda(n) \cdot (x - n). \quad (15)$$

By Lemma 3.6, we have

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{z+1}}{z(z+1)} \left(-\frac{\zeta'(z)}{\zeta(z)} \right) dz = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{z+1}}{z(z+1)} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^z} dz.$$

Set $a = \frac{x}{n} \geq 1$, then by applying Lemma 3.19, we have

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{z+1}}{z(z+1)} \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^z} dz = x \sum_{n=1}^{\infty} \Lambda(n) \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\left(\frac{x}{n}\right)^z}{z(z+1)} dz = x \sum_{n=1}^{\infty} \Lambda(n) \cdot \left(1 - \frac{n}{x}\right).$$

Then, by rewriting the above, we can apply (15) to obtain

$$x \sum_{n=1}^{\infty} \Lambda(n) \cdot \left(1 - \frac{n}{x}\right) = \sum_{n=1}^{\infty} \Lambda(n) \cdot (x - n) = \psi_1(x),$$

as required. $\square$

3.5 Estimations for $\frac{1}{\zeta}$ and ζ'

We need bounds for the rate of growth of $\frac{1}{\zeta}$ and ζ' . We first state a proposition without proof, taken from Stein and Shakarchi [17, C6. Proposition 2.7].

Proposition 3.21. *Suppose $z = \sigma + it$ with $\sigma, t \in \mathbb{R}$. Then, for each σ_0 such that $0 \leq \sigma_0 \leq 1$, and every $\varepsilon > 0$, there exists a constant c_ε so that*

$$(i) \quad |\zeta(z)| \leq c_\varepsilon |t|^{1-\sigma_0+\varepsilon}, \text{ if } \sigma_0 \leq \sigma \text{ and } |t| \geq 1,$$

$$(ii) \quad |\zeta'(z)| \leq c_\varepsilon |t|^\varepsilon, \text{ if } 1 \leq \sigma \text{ and } |t| \geq 1.$$

Now, we proceed to prove a result concerning the estimation of $\frac{1}{|\zeta(z)|}$, as stated in Stein and Shakarchi [17, C7. Proposition 1.6].

Proposition 3.22. *For every $\varepsilon > 0$, we have $\frac{1}{|\zeta(z)|} \leq c_\varepsilon |t|^\varepsilon$ where $z = \sigma + it$ with $\sigma \geq 1$ and $|t| \geq 1$.*

Proof. Let $\varepsilon > 0$. From Corollary 3.5, for $\sigma \geq 1$, we have

$$|\zeta^3(\sigma) \cdot \zeta^4(\sigma + it) \cdot \zeta(\sigma + 2it)| \geq 1. \quad (16)$$

Using our estimate for $|\zeta(z)|$ in Proposition 3.21, which states $|\zeta(z)| \leq c|t|^{1-\sigma_0+\varepsilon}$ for some $c \in \mathbb{R}$ with $0 \leq \sigma_0 \leq 1$, we choose $\sigma_0 = 1 \leq \sigma$ and subsequently have $|\zeta(z)| \leq c|t|^\varepsilon$. Applying this to (16) yields,

$$|\zeta^3(\sigma)| \cdot |\zeta^4(\sigma + it)| \cdot c|t|^\varepsilon \geq 1.$$

Then rearranging for $|\zeta^4(\sigma)|$ yields,

$$|\zeta^4(\sigma + it)| \geq c|t|^{-\varepsilon} \cdot |\zeta^{-3}(\sigma)| \geq c_1|t|^{-\varepsilon} \cdot (\sigma - 1)^3,$$

where the last inequality comes from the simple pole of ζ at $z = 1$. The constant c_1 combines c and any constant from the simple pole bound. The above holds for all $\sigma \geq 1$ and $|t| \geq 1$, hence

$$|\zeta(\sigma + it)| \geq c_2(\sigma - 1)^{\frac{3}{4}} \cdot |t|^{-\frac{\varepsilon}{4}}, \quad (17)$$

for $\sigma \geq 1$ and $|t| \geq 1$, and with $c_2 := c_1^{\frac{4}{3}}$.

We consider two cases. Suppose $\sigma - 1 \geq A|t|^{-5\varepsilon}$ holds for some $A \in \mathbb{R}$ which we choose later. Then from (17), we have

$$|\zeta(\sigma + it)| \geq c_2 A^{\frac{3}{4}} \cdot |t|^{-\frac{15\varepsilon}{4}} \cdot |t|^{-\frac{\varepsilon}{4}} = B \cdot |t|^{-4\varepsilon},$$

where $B := c_2 A^{\frac{3}{4}}$. So by replacing 4ε with ε , we can rearrange to get $\frac{1}{|\zeta(z)|} \leq B'|t|^\varepsilon$, with $B' = B^{-1}$, as required.

Alternatively, suppose $\sigma - 1 < A|t|^{-5\varepsilon}$. Then we first select some $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\varepsilon}$, which is possible due to the completeness of $\mathbb{R}$. Then, by the reverse triangle inequality, we have

$$|\zeta(\sigma + it)| \geq |\zeta(\sigma' + it)| - |\zeta(\sigma' + it) - \zeta(\sigma + it)|. \quad (18)$$

We consider each component of the left-hand side individually. By the mean value theorem, we have

$$|\zeta(\sigma' + it) - \zeta(\sigma + it)| \leq |\zeta'(z^*)(\sigma' - \sigma)|.$$

By Proposition 3.21, which states for $\sigma \geq 1$ that $|\zeta'(z)| \leq c_3|t|^\varepsilon$ for some $c_3 \in \mathbb{R}$, we have

$$|\zeta'(z^*)(\sigma' - \sigma)| = |\zeta'(z^*)| \cdot |\sigma' - \sigma| \leq c_3 \cdot |t|^\varepsilon \cdot |\sigma' - \sigma| \leq c_3 \cdot |t|^\varepsilon \cdot |\sigma' - 1|, \quad (19)$$

where the last inequality arises from $\sigma \geq 1$. Hence, by setting $\sigma = \sigma'$, we can utilise inequalities (18) and (19) in conjunction with (17) to obtain

$$|\zeta(\sigma + it)| \geq c_2(\sigma' - 1)^{\frac{3}{4}} \cdot |t|^{-\frac{\varepsilon}{4}} - c_3|\sigma' - 1| \cdot |t|^\varepsilon. \quad (20)$$

Choose $A = \left(\frac{c_2}{2c_3}\right)^4$ and recall $\sigma' - 1 = A|t|^{-5\varepsilon}$. It follows that,

$$c_2(\sigma' - 1)^{\frac{3}{4}} \cdot |t|^{-\frac{\varepsilon}{4}} = c_2 A^{\frac{3}{4}} \cdot |t|^{-4\varepsilon} = c_2 \left(\frac{c_2}{2c_3}\right)^3 \cdot |t|^{-4\varepsilon} = 2c_3 \cdot A \cdot |t|^{-4\varepsilon} = 2c_3(\sigma' - 1) \cdot |t|^\varepsilon.$$

Therefore, substituting the above into (20) yields

$$|\zeta(\sigma + it)| \geq 2c_3(\sigma' - 1) \cdot |t|^\varepsilon - c_3(\sigma' - 1) \cdot |t|^\varepsilon = c_3(\sigma' - 1) \cdot |t|^\varepsilon.$$

Then, using $\sigma' - 1 = A|t|^{-5\varepsilon}$, we have

$$c_3(\sigma' - 1) \cdot |t|^\varepsilon = c_3 \cdot A \cdot |t|^{-4\varepsilon},$$

so by replacing 4ε with ε , we again get the desired $\frac{1}{|\zeta(z)|} \leq c_3|t|^\varepsilon$, as required. This completes the proof. $\square$

Corollary 3.23. *For every $\varepsilon > 0$, we have $\left| \frac{\zeta'(z)}{\zeta(z)} \right| \leq c_\varepsilon |t|^\varepsilon$ where $z = \sigma + it$ with $\sigma \geq 1$ and $|t| \geq 1$.*

Proof. This directly follows from Proposition 3.22 and part (ii) of Proposition 3.21. $\square$

3.6 Prime Number Theorem

Recall in section 3.3 we proved the PNT based on an assumption, namely $\psi(x) \sim x$. In this subsection we prove this assumption holds unconditionally thus prove the PNT in its entirety. The following two theorems make use of our refined key ingredient, Proposition 3.20, our essential tool from complex analysis and some important estimations proved in the previous subsection. We have reformulated the discussion in Stein and Shakarchi [17, C7. Section 2.1] into the two subsequent theorems.

Theorem 3.24. *We unconditionally have, $\psi_1(x) \sim \frac{x^2}{2}$.*

Proof. From our refined key ingredient in Proposition 3.20, we have for $c > 1$

$$\psi_1(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{x^{z+1}}{z(z+1)} \left(-\frac{\zeta'(z)}{\zeta(z)} \right) dz,$$

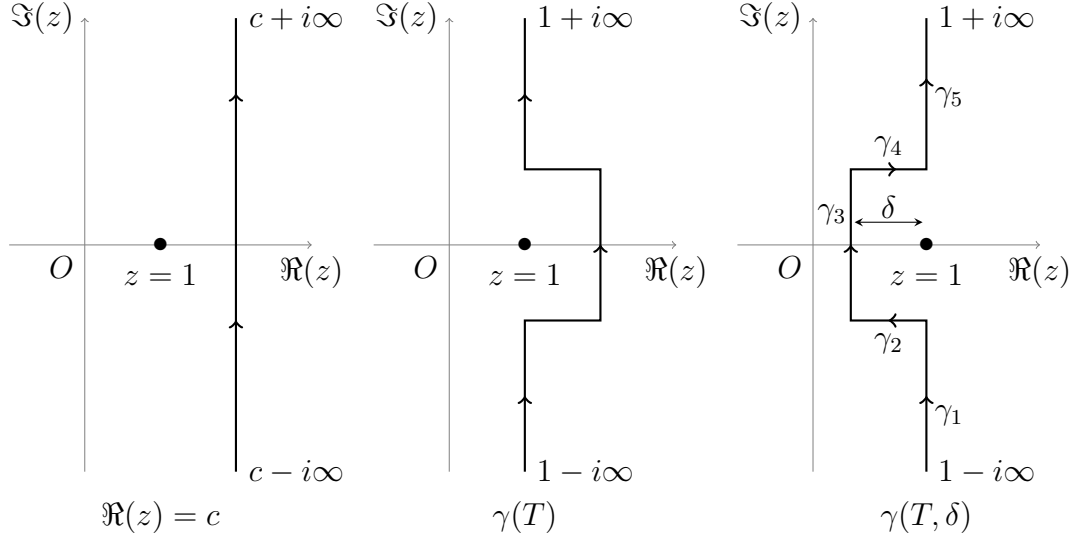
where the integral is taken on the line $\Re(z) = 1$. We will now use Cauchy's deformation theorem from complex analysis to transform this to integrate over the vertical line, $\Re(z) = 1$. Let us assume $x \geq 2$ and denote the integrand by

$$F(z) := \frac{x^{z+1}}{z(z+1)} \left(-\frac{\zeta'(z)}{\zeta(z)} \right).$$

We employ a sequence of deformations as outlined in Figure (2). We transition from the vertical line $\Re(z) = c$ to the paths $\gamma(T)$ and $\gamma(T, \delta)$. The value of T will be selected to be sufficiently large later.

Since the contour $\gamma(T)$ circumvents any singularities, by an application of Cauchy's deformation theorem, we have

$$\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(z) dz = \frac{1}{2\pi i} \int_{\gamma(T)} F(z) dz.$$

Figure 2: The contour denoted as $\Gamma(T)$

Subsequently, we transform the contour $\gamma(T)$ to the contour $\gamma(T, \delta)$. Given a fixed $T > 0$, we select $\delta > 0$ to be sufficiently small such that ζ has no zeros within the box given by

$$\{z \in \mathbb{C} : 1 - \delta \leq \Re(z) \leq 1, \text{ and } |\Im(z)| \leq T\}.$$

Such a choice exists due to Theorem 3.12, which asserts that ζ does not vanish on $\Re(z) = 1$. Since $F(z)$ has a pole of order 1 at $z = 1$, and we have $\text{Res}_{z=1} F(z) = \frac{x^2}{2}$, an application of the Residue theorem yields,

$$\frac{1}{2\pi i} \int_{\gamma(T)} F(z) dz = \frac{x^2}{2} + \frac{1}{2\pi i} \int_{\gamma(T, \delta)} \frac{x^{z+1}}{z(z+1)} F(z) dz. \quad (21)$$

We can now decompose the contour $\gamma(T, \delta)$ into $\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4 + \gamma_5$ as shown in Figure (2), and estimate each of the integral components $\int_{\gamma_j} F(z) dz$ for $j = 1, 2, 3, 4, 5$.

Firstly, we claim there exists $T \gg 1$ such that

$$\left| \int_{\gamma_1} F(z) dz \right| \leq \frac{\varepsilon}{2} x^2, \text{ and } \left| \int_{\gamma_5} F(z) dz \right| \leq \frac{\varepsilon}{2} x^2.$$

To observe this, let $z := 1 + it \in \gamma_1$. From the definition of $F(z)$, we have

$$\left| \int_{\gamma_1} F(z) dz \right| = \left| \int_{\gamma_1} \frac{x^{z+1}}{z(z+1)} \left(-\frac{\zeta'(z)}{\zeta(z)} \right) dz \right| \leq \int_{\gamma_1} \frac{|x^{z+1}|}{|z(z+1)|} \left| \frac{\zeta'(z)}{\zeta(z)} \right| |dz|.$$

Then, using $|t| \leq |z|$ and $|x^{1+z}| = x^{1+\sigma} = x^2$, we have

$$\int_{\gamma_1} \frac{|x^{z+1}|}{|z(z+1)|} \left| \frac{\zeta'(z)}{\zeta(z)} \right| |dz| \leq \int_{\gamma_1} \frac{x^2}{t^2} \left| \frac{\zeta'(z)}{\zeta(z)} \right| dt.$$

Then, by the asymptotics of the zeta function given in Corollary 3.23, we have $\left| \frac{\zeta'(z)}{\zeta(z)} \right| \leq C|t|^{\frac{1}{2}}$, hence

$$\int_{\gamma_1} \frac{x^2}{t^2} \left| \frac{\zeta'(z)}{\zeta(z)} \right| dt \leq Cx^2 \int_T^\infty \frac{|t|^{\frac{1}{2}}}{t^2} dt. \quad (22)$$

For $T > 0$, the above integral converges by Cauchy's integral convergence test as the corresponding summation $\sum_{n=T}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ converges by the p-series test. Since T is independent of x , by taking $T \rightarrow \infty$, we can make the expression in (22) arbitrarily small. Hence, we obtain the desired inequality $\left| \int_{\gamma_1} F(z) dz \right| \leq \frac{\varepsilon}{2} x^2$. A similar argument shows the same for $\left| \int_{\gamma_5} F(z) dz \right| \leq \frac{\varepsilon}{2} x^2$.

Given an arbitrary $\varepsilon > 0$, let us fix T so that the above condition is satisfied, and choose $0 < \delta \ll 1$ to satisfy the condition as well. On γ_3 , we have $|x^{z+1}| = x^{1+1-\delta} = x^{2-\delta}$, allowing us to find C_T such that

$$\left| \int_{\gamma_3} F(z) dz \right| \leq C_T \cdot x^{2-\delta}.$$

Finally, on the horizontal segment γ_2 , (and similarly on γ_4), we can estimate the corresponding integral as follows,

$$\left| \int_{\gamma_2} F(z) dz \right| \leq C'_T \int_{1-\delta}^1 x^{1+y} dy \leq C'_T \cdot \frac{x^2}{\log x},$$

where the final inequality makes use of

$$\int_{1-a}^1 x^{1+y} dy = \frac{x^{2-a}(x^a - 1)}{\log x}.$$

Thus, we can deduce the existence of constants C_T and C'_T that provide estimates for the integrals over γ_j . Consequently, we can now assemble each component and derive the following conclusion,

$$\left| \psi_1(x) - \frac{x^2}{2} \right| = \left| \int_{c-i\infty}^{c+i\infty} \frac{x^{z+1}}{z(z+1)} \left(-\frac{\zeta'(z)}{\zeta(z)} \right) dz - \frac{x^2}{2} \right|.$$

Then using the definition of $F(z)$, we have

$$\left| \int_{c-i\infty}^{c+i\infty} \frac{x^{z+1}}{z(z+1)} \left(-\frac{\zeta'(z)}{\zeta(z)} \right) dz - \frac{x^2}{2} \right| = \left| \int_{c-i\infty}^{c+i\infty} F(z) dz - \frac{x^2}{2} \right|.$$

Therefore, using (21), we have

$$\left| \int_{c-i\infty}^{c+i\infty} F(z) dz - \frac{x^2}{2} \right| = \left| \frac{x^2}{2} + \int_{\gamma(T,\delta)} \frac{x^{z+1}}{z(z+1)} F(z) dz - \frac{x^2}{2} \right|.$$

Hence,

$$\begin{aligned} \left| \frac{x^2}{2} + \int_{\gamma(T,\delta)} \frac{x^{z+1}}{z(z+1)} F(z) dz - \frac{x^2}{2} \right| &= \left| \int_{\gamma(T,\delta)} \frac{x^{z+1}}{z(z+1)} F(z) dz \right| \\ &\leq \frac{\varepsilon}{2} x^2 + \frac{\varepsilon}{2} x^2 + C_T x^{2-\delta} + 2C'_T \cdot \frac{x^2}{\log x}, \end{aligned}$$

where the final equality comes from using our integral estimates in the preceding parts of the proof. So we have shown,

$$\left| \psi_1(x) - \frac{x^2}{2} \right| \leq \frac{\varepsilon}{2} x^2 + \frac{\varepsilon}{2} x^2 + C_T x^{2-\delta} + 2C'_T \cdot \frac{x^2}{\log x}.$$

Dividing through by $\frac{x^2}{2}$ yields,

$$\left| \frac{2\psi_1(x)}{x^2} - 1 \right| \leq 2\varepsilon + 2C_T \frac{1}{x^\delta} + 4C'_T \frac{1}{\log x},$$

and taking $x \rightarrow \infty$ further yields,

$$\lim_{x \rightarrow \infty} \left| \frac{2\psi_1(x)}{x^2} - 1 \right| \leq 2\varepsilon.$$

Therefore, it follows

$$\lim_{x \rightarrow \infty} \left| \frac{\psi_1(x)}{\frac{x^2}{2}} \right| = 1,$$

which by definition, is equivalent to $\psi_1(x) \sim \frac{x^2}{2}$ thereby completing our proof. $\square$

Theorem 3.25. *If $\psi_1(x) \sim \frac{x^2}{2}$, then $\psi(x) \sim x$.*

Proof. If $\alpha < 1 < \beta$, then since ψ is an increasing function by definition, we have

$$\frac{1}{(1-\alpha)x} \int_{\alpha x}^x \psi(u) \, du \leq \psi(x) \leq \frac{1}{(\beta-1)x} \int_x^{\beta x} \psi(u) \, du.$$

So from the definition of ψ_1 , we have

$$\psi(x) \leq \frac{1}{(\beta-1)x} [\psi_1(\beta x) - \psi_1(x)].$$

Therefore,

$$\frac{\psi(x)}{x} \leq \frac{1}{(\beta-1)} \left[\frac{\psi_1(\beta x)}{(\beta x)^2} \beta^2 - \frac{\psi_1(x)}{x^2} \right].$$

So this implies that

$$\limsup_{x \rightarrow \infty} \frac{\psi(x)}{x} \leq \frac{1}{\beta-1} \left[\frac{1}{2} \beta^2 - \frac{1}{2} \right] = \frac{1}{2} \cdot \frac{\beta^2 - 1}{\beta - 1} = \frac{1}{2}(\beta + 1),$$

and since the result is true for all $\beta > 1$ we can take $\beta \rightarrow 1$ to obtain

$$\limsup_{x \rightarrow \infty} \frac{\psi(x)}{x} \leq 1.$$

Similarly, with $\alpha < 1$ one can show that $\liminf_{x \rightarrow \infty} \frac{\psi(x)}{x} \geq 1$. Then the sandwich theorem implies

$$\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1,$$

which precisely says that $\psi(x) \sim x$. $\square$

From Theorem 3.24, we have established the following asymptotic relation: $\psi_1(x) \sim \frac{x^2}{2}$. Then, by applying Theorem 3.25, we have $\psi(x) \sim x$. Finally, from Theorem 3.17, we have $\pi(x) \sim \frac{x}{\log x}$, which constitutes the PNT. We state this more formally below.

Theorem 3.26 (Prime Number Theorem). *The PNT asserts the following asymptotic growth of the prime counting function, $\pi(x)$:*

$$\pi(x) \sim \frac{x}{\log x} \iff \lim_{x \rightarrow \infty} \frac{\pi(x) \log(x)}{x} = 1.$$

4 Order of magnitude of the divisor function

Recall, Chebychev's Theorem 2.1 showed the order of magnitude of $\pi(x)$ is $\frac{x}{\log x}$. In this standalone section, we illustrate how both the PNT and Chebychev's theorem can be applied to establish order of magnitude results for certain multiplicative arithmetic functions.

Definition 4.1. A multiplicative arithmetical function is a function $f : \mathbb{N} \rightarrow \mathbb{C}$ such that $f(1) = 1$ and $f(ab) = f(a)f(b)$.

Let $d(n) = \sum_{k|n} 1$ be the number of divisors for some $n \in \mathbb{N}$. Then $d(n)$ is a multiplicative arithmetical function and we will prove the bounds in Theorem 4.4 which is taken from Apostol [3, C13. Theorem 13.12]. At this point, it is useful to introduce big-O and little-o notation, which will be used in this section and section 6 of our essay.

Definition 4.2. Let f and g be two functions. We say $f(x)$ is big-O of $g(x)$ if there exists there exists $M, x_0 > 0$ such that

$$|f(x)| \leq Mg(x),$$

for all $x \geq x_0$. We denote this as $f(x) = \mathcal{O}(g(x))$.

Definition 4.3. Let f and g be two functions. We say $f(x)$ is little-o of $g(x)$ if for all $\varepsilon > 0$, there exists $x_0 > 0$ such that

$$|f(x)| \leq \varepsilon g(x),$$

for all $x \geq x_0$. We denote this as $f(x) = o(g(x))$.

Theorem 4.4. Let $\varepsilon > 0$ be given. Then

1. $\exists N_\varepsilon$ such that for all $n > N_\varepsilon$, we have

$$d(n) < 2^{\left\lfloor \frac{(1+\varepsilon)\log n}{\log \log n} \right\rfloor} = n^{\left\lfloor \frac{(1+\varepsilon)\log 2}{\log \log n} \right\rfloor}.$$

2. $\exists N'_\varepsilon$ such that for all $n > N'_\varepsilon$, we have

$$d(n) > 2^{\left\lfloor \frac{(1-\varepsilon)\log n}{\log \log n} \right\rfloor} = n^{\left\lfloor \frac{(1-\varepsilon)\log 2}{\log \log n} \right\rfloor}.$$

We follow the proof outlined in Apostol [3, Pg. 295], while including more detail and additional explanations for each step.

Proof. By the fundamental theorem of arithmetic, let $n = p_1^{a_1} \cdots p_k^{a_k}$ and let $f : \mathbb{N} \rightarrow \mathbb{C}$ be a function which will be defined later. Then $d(n) = \prod_{i=1}^k (a_i + 1)$ where $a_i + 1$ indicates the number of powers of p_i to include in some general divisor. We split this product into 2 parts, separating those divisors where every prime factor is less than $f(n)$ and those divisors where every prime factor is greater than or equal to $f(n)$. More precisely,

$$d(n) = P_1(n) \cdot P_2(n), \text{ where } P_1(n) = \prod_{\substack{i=1 \\ p_i < f(n)}}^k (a_i + 1), \quad P_2(n) = \prod_{\substack{i=1 \\ p_i \geq f(n)}}^k (a_i + 1).$$

For estimating the product $P_2(n)$, we make use of the inequality $(a+1) \leq 2^a$ for all $a \geq 1$ to obtain

$$P_2(n) = \prod_{\substack{i=1 \\ p_i \geq f(n)}}^k (a_i + 1) \leq \prod_{\substack{i=1 \\ p_i \geq f(n)}}^k 2^{a_i} = 2^{S(n)} \text{ where } S(n) := \sum_{\substack{i=1 \\ p_i \geq f(n)}}^k a_i. \quad (23)$$

Then, we have

$$n = \prod_{i=1}^k p_i^{a_i} \geq \prod_{\substack{i=1 \\ p_i \geq f(n)}}^k p_i^{a_i} \geq \prod_{\substack{i=1 \\ p_i \geq f(n)}}^k f(n)^{a_i} = f(n)^{S(n)}.$$

Hence, taking logarithms yields

$$\log(n) \geq S(n) \cdot \log f(n) \iff S(n) \leq \frac{\log n}{\log f(n)}. \quad (24)$$

Thus we can estimate $P_2(n)$ by using (23) and (24)

$$P_2(n) \leq 2^{S(n)} \leq 2^{\frac{\log n}{\log f(n)}}. \quad (25)$$

To estimate $P_1(n)$, we express it as

$$P_1(n) = \prod_{\substack{i=1 \\ p_i < f(n)}}^k (a_i + 1) = \exp \left\{ \log \prod_{\substack{i=1 \\ p_i < f(n)}}^k (a_i + 1) \right\} = \exp \left\{ \sum_{\substack{i=1 \\ p_i < f(n)}}^k \log(a_i + 1) \right\}. \quad (26)$$

Our goal is to bound the quantity $\log(a_i + 1)$ and we claim $\log(a_i + 1) < 2 \log \log n$ for sufficiently large n . Fix some i , then $n \geq p_i^{a_i} \geq 2^{a_i}$. Thus taking logarithms yields

$$\log n \geq a_i \cdot \log 2 \iff a_i \leq \frac{\log n}{\log 2}.$$

Therefore, since a quadratic grows faster than a linear term, there exists $n_1 \in \mathbb{N}$ such that we have

$$a_i + 1 \leq \frac{\log n}{\log 2} + 1 < (\log n)^2,$$

for all $n \geq n_1$. Therefore, by taking an additional logarithm, we have $\log(a_i + 1) < \log(\log n)^2 = 2 \log \log n$. We now apply this result to our expression in (26) to get

$$P_1(n) < \exp \left\{ 2 \log \log n \cdot \sum_{\substack{i=1 \\ p_i < f(n)}}^k 1 \right\} \leq \exp \{ 2(\log \log n) \cdot \pi(f(n)) \}.$$

Theorem 2.1 tells us that $\pi(x) \leq \frac{6x}{\log x}$ for $x \geq 2$, therefore

$$P_1(n) < \exp \left\{ \frac{12 \cdot f(n) \cdot \log \log n}{\log f(n)} \right\} = 2^{\frac{c \cdot f(n) \cdot \log \log n}{\log f(n)}}, \quad (27)$$

where $c = \frac{12}{\log 2}$. Now using our estimates for $P_1(n)$ and $P_2(n)$ as seen in (27) and (25), we have

$$d(n) = P_1(n)P_2(n) < 2^{\frac{\log n}{\log f(n)} + \frac{c \cdot f(n) \cdot \log \log n}{\log f(n)}} = 2^{g(n)},$$

where it is clear $g(n)$ is defined as the exponent of 2 in the above expression. We express $g(n)$ in a more useful form as follows

$$g(n) = \frac{\log n}{\log \log n} \cdot \left(\frac{\log \log n}{\log f(n)} + \frac{c \cdot f(n) \cdot (\log \log n)^2}{\log n \cdot \log f(n)} \right) = \frac{\log(n)}{\log \log n} \left(\frac{1}{\frac{\log f(n)}{\log \log n}} + \frac{\frac{c \cdot f(n) \cdot \log \log n}{\log n}}{\frac{\log f(n)}{\log \log n}} \right). \quad (28)$$

We require a choice of $f(n)$ such that $\frac{f(n) \cdot \log \log n}{\log n} \rightarrow 0$ and $\frac{\log f(n)}{\log \log n} \rightarrow 1$ as $n \rightarrow \infty$. Let

$$f(n) = \frac{\log n}{(\log \log n)^2}.$$

It can easily be verified $f(n)$ satisfies the aforementioned conditions. By substituting $f(n)$ into (28), we have

$$g(n) = \frac{\log n}{\log \log n} (1 + o(1)) < (1 + \varepsilon) \frac{\log n}{\log \log n},$$

for sufficiently large $n > N_\varepsilon$. This proves the upper bound. We have currently only made use of Chebychev's theorem. For the lower bound we will make use of the PNT as well as an equivalent form, as found in Apostol [3, C4. Theorem 4.4] or Theorem 5.4, which states $\vartheta(x) := \sum_{p \leq x} \log p \sim x$

Let n be an integer with a large number of prime factors, we set $n = \prod_{p \leq x} p$. Then we can choose whether to include a prime factor in a general divisor or not, therefore

$$d(n) = 2^{\pi(x)} = 2^{\frac{(1 + o(1))x}{\log x}}, \quad (29)$$

where the last equality is by the PNT. Also, for this n ,

$$\log n = \log \prod_{p \leq x} p = \sum_{p \leq x} \log p = \vartheta(x) = x(1 + o(1)),$$

where the final equality is by the aforementioned equivalent form of the PNT. So it follows

$$x = \frac{\log n}{1 + o(1)} = (1 + o(1)) \log n. \quad (30)$$

Hence,

$$\log x = \log \log n + \log(1 + o(1)) = \log \log n \left(1 + \frac{\log(1 + o(1))}{\log \log n} \right) = (1 + o(1)) \log \log n. \quad (31)$$

Therefore, we can replace $\frac{x}{\log x}$ in (29) by applying (30) and (31), to obtain

$$d(n) = 2^{\frac{(1 + o(1)) \log n}{\log \log n}},$$

for the specified n . Let $\varepsilon > 0$, then $1 + o(1) > 1 - \varepsilon$ so for sufficiently large n , the lower bound is also proven. $\square$

5 Stepping into results on gaps between consecutive primes

In this section, we begin by using an equivalent form of the PNT, namely $p_n \sim n \log n$, to showcase how this implies a conditional bound on $p_{n+1} - p_n$. Subsequently, we provide a survey of known results that improve upon this bound, and discuss the implications of assuming some well-known conjectures on bounding $p_{n+1} - p_n$. Additionally, we present primary computations that lead us to conjecture a result whose validity is implied by the Andrica conjecture. Furthermore, we delve into primes between consecutive squares and cubes, with the former being an implication of Andrica's conjecture and the latter being true for sufficiently large n . Finally, we provide a proof demonstrating that $\sqrt{p_{n+1}} - \sqrt{p_n} \rightarrow 0$ as $n \rightarrow \infty$, under the assumption of Cramér's conjecture.

5.1 A conditional bound on $p_{n+1} - p_n$ using the PNT

Previously, we demonstrated that $\psi(x) \sim x$ holds and further showed that this implies the PNT. In fact, these two statements are equivalent. In this section, we present a series of useful equivalent forms of the PNT, all of which we utilise in some form in this essay. Then, using one of these forms, we provide a conditional bound for $p_{n+1} - p_n$ as a consequence of the PNT.

Definition 5.1. Suppose $x \in \mathbb{N}$, then the *first Chebychev* function, $\vartheta(x)$, is defined as

$$\vartheta(x) = \sum_{p \leq x} \log p.$$

Remark 5.2. Recall the definition for the second Chebychev function in 3.13. We can express $\psi(x)$ in terms of $\vartheta(x)$ as follows

$$\psi(x) = \sum_{m=1}^{\infty} \sum_{p^m \leq x} \log p = \sum_{m=1}^{\infty} \sum_{p \leq x^{\frac{1}{m}}} \log p = \sum_{m=1}^{\infty} \vartheta(x^{\frac{1}{m}}).$$

However, $\vartheta(x^{\frac{1}{m}}) = 0$ for $m > \log_2(x)$, since $x^{\frac{1}{\log_2(x)}} = 2$. Therefore, $\psi(x) = \sum_{m \leq \log_2(x)} \vartheta(x^{\frac{1}{m}})$.

The following result relates the first Chebychev function to the second. We present the ideas found in Apostol [3, C4. Theorem 4.1] in a more explicit way.

Theorem 5.3. *We have,*

$$\lim_{x \rightarrow \infty} \left(\frac{\psi(x)}{x} - \frac{\vartheta(x)}{x} \right) = 0,$$

and consequently,

$$\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = \lim_{x \rightarrow \infty} \frac{\vartheta(x)}{x} = 1.$$

Proof. By Remark 5.2, we have

$$0 \leq \psi(x) - \vartheta(x) = \sum_{m=2}^{\log_2(x)} \vartheta(x^{\frac{1}{m}}). \quad (32)$$

Then, by the definition of $\vartheta(x)$, we have

$$\vartheta(x) = \sum_{p \leq x} \log p \leq \sum_{p \leq x} \log x \leq x \cdot \log x. \quad (33)$$

Thus, by combining (32) and (33), we obtain

$$0 \leq \psi(x) - \vartheta(x) \leq \sum_{m=2}^{\log_2(x)} x^{\frac{1}{m}} \cdot \log(x^{\frac{1}{m}}) \leq (\log_2(x)) \cdot \sqrt{x} \cdot \log(\sqrt{x}) = \frac{\log x}{\log 2} \cdot \frac{\sqrt{x}}{2} \cdot \log x = \frac{\sqrt{x}(\log x)^2}{2 \log 2}.$$

By dividing through by x , we have

$$0 \leq \frac{\psi(x)}{x} - \frac{\vartheta(x)}{x} \leq \frac{(\log x)^2}{2\sqrt{x} \log 2}.$$

By applying L'Hôpital's rule twice, we observe that

$$\lim_{x \rightarrow \infty} \frac{(\log x)^2}{2\sqrt{x} \log 2} = 0.$$

Then, by the sandwich theorem for limits, we establish our first claim. From Theorem 3.25, we have

$$\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1,$$

and combining this with our first claim completes the proof. $\square$

Theorem 5.4 (Equivalent representations of the PNT). *Using p_n to denote the n^{th} prime number, the PNT is equivalent to the following asymptotic relations*

- (i) $\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1,$
- (ii) $\lim_{x \rightarrow \infty} \frac{\vartheta(x)}{x} = 1,$
- (iii) $\lim_{n \rightarrow \infty} \frac{p_n}{n \log n} = 1.$

Proof. A complete proof of equivalence can be found in Apostol [3, C4. Theorem 4.5]. However, for our purposes, we will simply demonstrate the truth of each limit. We have already established the truth of (i) in Theorem 3.25. Theorem 5.3 establishes that (i) is equivalent to (ii). Now, we demonstrate the truth of (iii) using the PNT. We have $\pi(x) \sim \frac{x}{\log x}$ can be expressed as

$$\lim_{x \rightarrow \infty} \frac{\pi(x) \log x}{x} = 1.$$

By letting $x = p_n$ and observing that $\pi(p_n) = n$, we have

$$\lim_{n \rightarrow \infty} \frac{\pi(p_n) \log p_n}{p_n} = 1 \iff \lim_{n \rightarrow \infty} \frac{n \log p_n}{p_n} = 1. \quad (34)$$

Now, taking the logarithms of (34), we have

$$\lim_{n \rightarrow \infty} (\log n + \log \log p_n - \log p_n) = 0.$$

So it follows,

$$\lim_{n \rightarrow \infty} \log p_n \left(\frac{\log n}{\log p_n} + \frac{\log \log p_n}{\log p_n} - 1 \right) = 0.$$

However, $\log p_n \rightarrow \infty$ and $\frac{\log \log p_n}{\log p_n} \rightarrow 0$ as $n \rightarrow \infty$. Therefore,

$$\lim_{n \rightarrow \infty} \frac{\log n}{\log p_n} = 1. \quad (35)$$

So by multiplying (34) and (35) together, we obtain our desired result. $\square$

We use the representation of the PNT which states $p_n \sim n \log n$ to show a conditional bound on $p_{n+1} - p_n$.

Proposition 5.5. *For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$, we have*

$$p_{n+1} - p_n < \varepsilon \cdot p_n.$$

Proof. From $p_n \sim n \log n$, we have the following

$$\lim_{n \rightarrow \infty} \frac{p_n}{n \log n} = \lim_{n \rightarrow \infty} \frac{p_{n+1}}{(n+1) \log(n+1)} = 1.$$

Therefore,

$$\lim_{n \rightarrow \infty} \left(\frac{p_{n+1}}{(n+1) \log(n+1)} - \frac{p_n}{n \log n} \right) = 0. \quad (36)$$

Now, we apply L'Hôpital's rule twice

$$\lim_{n \rightarrow \infty} \frac{(n+1) \log(n+1)}{n \log n} = \lim_{n \rightarrow \infty} \frac{1 + \log(n+1)}{1 + \log n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n+1}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = 1,$$

so $(n+1) \log(n+1) \sim n \log n$. Applying this to (36) and using $p_n \sim n \log n$ yields

$$\lim_{n \rightarrow \infty} \frac{p_{n+1} - p_n}{n \log n} = \lim_{n \rightarrow \infty} \frac{p_{n+1} - p_n}{p_n} = 0,$$

and the result follows by using the epsilon definition of a limit. $\square$

5.2 A survey of improved unconditional and conditional bounds on

$$p_{n+1} - p_n$$

We have observed that $p_{n+1} - p_n < \varepsilon \cdot p_n$ for all $\varepsilon > 0$. A natural question arises: can we improve this bound? In the following section, we provide a survey of improved conditional and unconditional bounds.

The following series of results heavily rely on the zeros of the zeta function, $\zeta(s)$. In the absence of a proof for the Riemann Hypothesis, one must typically assume the presence of zeros of $\zeta(\sigma + it)$ within the range $\frac{1}{2} < \sigma < 1$, and then demonstrate that the impact of these

zeros is not significant. Let us define a function that measures the number of zeros in such a region by

$$N(\alpha, T) := \#\{s = \beta + i\gamma \mid \zeta(s) = 0, \alpha \leq \beta \leq 1, -T \leq \gamma \leq T\}.$$

Following the bound presented in Proposition 5.5, the next milestone was achieved by Hoheisel [9] who used known estimations on $N(\alpha, T)$, known as zero density estimates for the zeta function.

Theorem 5.6 (Hoheisel, 1930). *Let $\theta = \frac{32999}{33000}$. Then, there exists $N \in \mathbb{N}$ such that*

$$p_{n+1} - p_n < p_n^\theta \quad \text{for all } n \geq N.$$

Six years later, Tchudakoff [19] extended upon this result by showing that it applies to all $\theta > 0.75$. He achieved this by improving a zero density estimate used by Hoheisel. In his original paper, Hoheisel noted that the value of $\theta > 0.75$ could be attained if such a modification was feasible.

Theorem 5.7 (Chudakov, 1936). *For all $\theta > 0.75$, there exists $N \in \mathbb{N}$ such that*

$$p_{n+1} - p_n < p_n^\theta \quad \text{for all } n \geq N. \quad (37)$$

The work of Huxley [10] and Montgomery used better estimations of $N(\alpha, T)$ to improve θ in (37).

Theorem 5.8 (Montgomery, 1969). *For all $\theta > 0.6$, there exists $N \in \mathbb{N}$ such that (37) holds for all $n \geq N$.*

The above bound was refined by Huxley [10] by realising slight improvements to Montgomery's zero density estimate.

Theorem 5.9 (Huxley, 1971). *For all $\theta > \frac{7}{12}$, there exists $N \in \mathbb{N}$ such that (37) holds for all $n \geq N$.*

The most recent result is that of Baker et al. [4] who no longer use zero density estimates but rather, mean value results on Dirichlet polynomials alongside a two-dimensional sieve to achieve the following.

Theorem 5.10 (Baker, Harman, Pintz, 2001). *For all $\theta > 0.525$, there exists $N \in \mathbb{N}$ such that (37) holds for all $n \geq N$.*

The next significant breakthrough was achieved by Ingham [11], just a year after Chudakov, who demonstrated an implication between the Lindelöf hypothesis and $p_{n+1} - p_n < p_n^\theta$.

Conjecture 5.11 (Lindelöf hypothesis). *For every $c > 0$, we have*

$$\zeta\left(\frac{1}{2} + it\right) = \mathcal{O}(t^c).$$

Theorem 5.12 (Ingham, 1937). *Suppose the Lindelöf hypothesis (5.11), holds for some $c > 0$. Then for any $\theta > \frac{1+4c}{2+4c}$, there exists $N \in \mathbb{N}$ such that*

$$p_{n+1} - p_n < p_n^\theta \quad \text{for all } n \geq N. \quad (38)$$

We observe that if the Lindelöf hypothesis were true, then θ could be reduced to 0.5. Known values of c for which Conjecture 5.11 holds include that of Hardy and Littlewood [7], who showed $c = \frac{1}{6}$. We will make use of this result to prove the existence of primes between sufficiently large consecutive cubes as seen in Theorem 5.17.

One may inquire, for which θ does the inequality in (38) hold for all $n \in \mathbb{N}$? Based on computational data (see Appendix C), we conjecture the following.

Conjecture 5.13. *For $\theta = \frac{\log 4}{\log 7} \approx 0.7125$, we have*

$$p_{n+1} - p_n < p_n^\theta < p_n^\theta + \frac{1}{4}.$$

The latter inequality is more useful to demonstrate that our conjecture is implied by an existing conjecture, as shown in the next section. Furthermore, as mentioned in Baker et al. [4], by investing sufficient effort, the value of N could be ascertained by adapting the entire proof to incorporate explicit values. However, doing this manually would take a vast amount of computational time which is currently not feasible. Given that our proposed value of $\theta = \frac{\log 4}{\log 7}$ surpasses that in Theorem 5.10, we could compute the finite number of cases up to this explicitly determined N , thus substantiating our conjecture.

5.3 A connection to the Andrica conjecture

We begin by introducing a different, well-known conjecture by Dorin Andrica [2] and then relating this to our previously proposed, Conjecture 5.13.

Conjecture 5.14 (Andrica, 1986). *For all $n \in \mathbb{N}$, we have*

$$\sqrt{p_{n+1}} - \sqrt{p_n} < 1.$$

We also cite a stronger version of the Andrica conjecture from Visser [20]:

Conjecture 5.15 (Strong Andrica, 2019). *Except for $p_n \in \{3, 7, 13, 23, 31, 113\}$ that is, $n \in \{2, 4, 6, 9, 11, 30\}$, we have:*

$$\sqrt{p_{n+1}} - \sqrt{p_n} < \frac{1}{2}. \tag{39}$$

We demonstrate how Conjecture 5.15 implies our Conjecture 5.13. Squaring both sides of (39) yields,

$$p_{n+1} - 2\sqrt{p_n}\sqrt{p_{n+1}} + p_n < \frac{1}{4}.$$

Hence, under the assumption of Conjecture 5.15, we have

$$p_{n+1} + p_n < 2\sqrt{p_n}\sqrt{p_{n+1}} + \frac{1}{4} < 2\sqrt{p_n}(\sqrt{p_n} + \frac{1}{2}) + \frac{1}{4} = 2p_n + p_n^{\frac{1}{2}} + \frac{1}{4}.$$

Therefore,

$$p_{n+1} - p_n < p_n^{\frac{1}{2}} + \frac{1}{4}.$$

Comparing this to our Conjecture 5.13, we have

$$p_{n+1} - p_n < p_n^{\frac{1}{2}} + \frac{1}{4} < p_n^{0.7125} + \frac{1}{4} \quad \text{with } n \geq 31.$$

So by assuming the strong Andrica conjecture and computing a finite number of cases, $n \leq 30$, we can show the second inequality of Conjecture 5.13 holds. Given the Strong Andrica conjecture has been verified for all primes $p < 2^{64} \approx 1.844 \times 10^{19}$, as seen in Vissers paper, this makes our Conjecture 5.13 plausible to hold.

5.4 Andrica's conjecture implies a weaker conjecture from Legendre

Conjecture 5.16 (Legendre's Conjecture). *For every positive integer, m , there exists a prime number, p such that $m^2 < p < (m+1)^2$.*

By assuming the Andrica conjecture, we can provide a proof to Legendre's conjecture in the following way.

Proof. From Andrica's conjecture, we have

$$(\sqrt{p_{n+1}} - \sqrt{p_n})^2 < 1.$$

By expanding and rearranging terms, we yield

$$p_{n+1} - p_n < 1 - 2p_n + 2\sqrt{p_{n+1}}\sqrt{p_n} = 1 + 2\sqrt{p_n}(\sqrt{p_{n+1}} - \sqrt{p_n}) < 1 + 2\sqrt{p_n}.$$

Let $m \in \mathbb{N}$ such that $m > \max\{2, p_n\}$ where p_n is the greatest prime number smaller than or equal to m^2 . Then

$$m^2 < p_{n+1} < 2\sqrt{p_n} + 1 + p_n = (\sqrt{p_n} + 1)^2 < (\sqrt{m^2} + 1)^2 = (m+1)^2, \quad (40)$$

thereby proving Legendre's conjecture. $\square$

As mentioned before, we can make use of Ingham's Theorem 5.12, and a partial result to Conjecture 5.11 by Hardy and Littlewood [7] who showed the conjecture holds for $c = \frac{1}{6}$, to provide a proof to the following theorem.

Theorem 5.17 (Primes between consecutive cubes). *There exists $n_1 \in \mathbb{N}$ such that for all $n \geq n_1$, there exists a prime p satisfying $p \in (n^3, (n+1)^3)$.*

Proof. Let $\varepsilon > 0$ which we choose later. Using Theorem 5.12 and the result by Hardy and Littlewood [7], we have there exists $n_0 \in \mathbb{N}$ such that

$$p_{n+1} - p_n < p_n^{\theta+\varepsilon} \quad \text{for all } n \geq n_0, \text{ where } \theta = \frac{1+4c}{2+4c} = \frac{5}{8}. \quad (41)$$

Choose $N > n_0$ sufficiently large such that there exists a prime p_m with $m \geq n_0$ with the property, p_m is the greatest prime satisfying $p_m \leq N^3$. Since $m \geq n_0$, we can apply (41) to obtain

$$N^3 < p_{m+1} < p_m^{\frac{5}{8}+\varepsilon} + p_m < N^{\frac{15}{8}+3\varepsilon} + N^3.$$

Choose ε such that $\frac{15}{8} + 3\varepsilon < 2$, then it follows

$$N^{\frac{15}{8}+3\varepsilon} + N^3 < N^3 + N^2 < N^3 + 3N^2 + 3N + 1 = (N+1)^3.$$

Hence,

$$N^3 < p_{m+1} < (N+1)^3.$$

Therefore, we have showed $p_{m+1} \in (N^3, (N+1)^3)$, as required. $\square$

5.5 Cramér's conjecture

We've observed that under the assumption of the Lindelöf hypothesis, we can reduce θ in (38) to 0.5. Now, we introduce a much stronger conjecture from Cramér, which implies that θ can be any positive real number. As discussed in Pintz [16], Cramér formulated a probabilistic model for the primes using the PNT. From this model, he derived a profound conjecture concerning large differences between consecutive primes. In this section, we outline how his probabilistic model motivates this conjecture and demonstrate how it allows θ to take on any positive real value. Additionally, we offer a concise proof, assuming Cramér's conjecture, that $\sqrt{p_{n+1}} - \sqrt{p_n} \rightarrow 0$ as $n \rightarrow \infty$. We make use of the formulation in Pintz [16, (1.2)].

Theorem 5.18 (Cramér's Probabilistic Model). *For $n \geq 3$, define a sequence of independent random variables, $\xi(n)$, by*

$$\mathbb{P}(\xi(n) = 1) = \frac{1}{\log n}, \quad \mathbb{P}(\xi(n) = 0) = 1 - \frac{1}{\log n}.$$

Then, we have

$$\mathbb{P}_{(\xi(n))_{n \geq 3}} \left[\limsup_{n \rightarrow \infty} \frac{\xi(n+1) - \xi(n)}{(\log \xi(n))^2} = 1 \right] = 1.$$

Note that the definition of $\xi(n)$ aligns with the PNT, which asserts that the probability of n being prime, within the range $[1, x]$, is $\frac{1}{\log x}$. Note that Theorem 5.18 is a probabilistic result. Therefore, it serves as a heuristic. From this, Cramér conjectures the following.

Conjecture 5.19 (Cramér's Conjecture). *We have,*

$$\limsup_{n \rightarrow \infty} \frac{p_{n+1} - p_n}{(\log p_n)^2} = 1.$$

We now show how Cramér's conjecture allows us to take θ in (38) to be any positive real number.

Corollary 5.20. *Assuming Cramér's Conjecture 5.19, for all $\theta > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have*

$$p_{n+1} - p_n < p_n^\theta.$$

Proof. Let $\varepsilon = \frac{1}{m^2} - 1$. From Conjecture 5.19, there exists $N \in \mathbb{N}$ such that for all $n > N$, we have

$$\frac{p_{n+1} - p_n}{(\log p_n)^2} < 1 + \varepsilon \iff p_{n+1} - p_n < (1 + \varepsilon) \cdot (\log p_n)^2. \quad (42)$$

We know $x \geq \log x$ for all $x > 0$, therefore $x^{\frac{1}{m}} \geq \log x^{\frac{1}{m}}$ for all $m > 1$. Hence, $mx^{\frac{1}{m}} \geq \log x$, for all $m > 1$. Applying this to (42) yields

$$p_{n+1} - p_n < (1 + \varepsilon) \cdot (m \cdot p_n^{\frac{1}{m}})^2 = (1 + \varepsilon) \cdot m^2 \cdot p_n^{\frac{2}{m}}, \quad \text{for all } m > 1. \quad (43)$$

From our choice of ε , we have $(1 + \varepsilon)m^2 = 1$ and thus (43) reduces to

$$p_{n+1} - p_n < p_n^{\frac{2}{m}}.$$

Since our choice of m is arbitrary, this proves our claim. $\square$

As a consequence to Corollary 5.20, we can now easily show $\sqrt{p_{n+1}} - \sqrt{p_n} \rightarrow 0$ as $n \rightarrow \infty$.

Theorem 5.21. *We have,*

$$\lim_{n \rightarrow \infty} (\sqrt{p_{n+1}} - \sqrt{p_n}) = 0.$$

Proof. Let $\varepsilon > 0$ and fix some $k \in (0, \frac{1}{2})$. Then

$$\lim_{n \rightarrow \infty} \frac{p_n^k}{(\sqrt{p_n} + \varepsilon)^2 - p_n} = \lim_{n \rightarrow \infty} \frac{p_n^k}{2\varepsilon\sqrt{p_n} + \varepsilon^2} = 0,$$

where the last equality comes from $k < \frac{1}{2}$. So it follows there exists $N_1 \in \mathbb{N}$ such that for $n > N_1$,

$$p_n^k < (\sqrt{p_n} + \varepsilon)^2 - p_n.$$

Therefore,

$$p_n + p_n^k < (\sqrt{p_n} + \varepsilon)^2. \quad (44)$$

From Corollary 5.20, we have there exists $N_2 \in \mathbb{N}$ such that for $n > N_2$, we have $p_{n+1} < p_n + p_n^k$. Set $N = \max\{N_1, N_2\}$. Then combined with (44), for $n > N$ we have

$$p_{n+1} < p_n + p_n^k < (\sqrt{p_n} + \varepsilon)^2.$$

Taking square roots and rearranging yields,

$$\sqrt{p_{n+1}} - \sqrt{p_n} < \varepsilon,$$

which is equivalent to $\lim_{n \rightarrow \infty} (\sqrt{p_{n+1}} - \sqrt{p_n}) = 0$, as required. $\square$

It is worth mentioning that the outcome of Theorem 5.21 almost corroborates the Andrica conjecture as posited in Conjecture 5.14. By setting $\varepsilon = 1$, we can apply Theorem 5.21 to establish the existence of $N \in \mathbb{N}$ such that $\sqrt{p_{n+1}} - \sqrt{p_n} < 1$. If there were a method to explicitly determine this value of N , then by validating Andrica's conjecture for the remaining cases below N , we could effectively prove Andrica's conjecture.

Hence, through the amalgamation of Corollary 5.20, Theorem 5.21, and the preceding discussion, we have essentially furnished a nearly comprehensive proof that Cramér's conjecture implies the Andrica conjecture.

6 An Elementary Proof

In 1921, Hardy presented a lecture to the Mathematical Society of Copenhagen in which he says the following.

No elementary proof of the prime number theorem is known, and one may ask whether it is reasonable to expect one. Now we know that the theorem is roughly equivalent to a theorem about an analytic function, the theorem that Riemann's zeta function has no roots on a certain line. A proof of such a theorem, not fundamentally dependent on the theory of functions, seems to me extraordinarily unlikely. It is rash to assert that a mathematical theorem cannot be proved in a particular way; but one thing seems quite clear. We have certain views about the logic of the theory; we think that some theorems, as we say 'lie deep' and others nearer to the surface. If anyone produces an elementary proof of the prime number theorem, he will show that these views are wrong, that the subject does not hang together in the way we have supposed, and that it is time for the books to be cast aside and for the theory to be rewritten.

— G. H. HARDY, 1921

The excerpt is sourced from Goldfeld [6], which offers a historical narrative of the events related to the PNT. Contrary to Hardy's beliefs, Atle Selberg and Paul Erdős collaborated to offer an *elementary* proof of the PNT, published in 1949.

6.1 Selberg-Erdős' elementary proof of the Prime Number Theorem

Recall from Remark 3.14, we showed $\psi(x) = \sum_{n \leq x} \Lambda(n)$ where $\Lambda(n)$ is the previously defined Von Mangoldt function. In Theorem 5.4, we showed the following limit is an equivalent result to the PNT.

$$\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1. \quad (45)$$

This elementary approach establishes (45) by employing a sequence of manipulations to a well-known result of Selberg. It should be stressed that *elementary* does not mean the manipulations are trivial or simple; in fact they can be very involved and technical. Elementary means we do not utilise any advanced machinery such as complex analysis as seen in chapter 3.

This section is primarily inspired by Levinson [15], where all proofs are provided. For simplicity, we omit any proofs involving intricate deductions and focus solely on elucidating the critical steps necessary to achieve our desired result in (45).

We begin with a major result first derived by Atle Selberg in 1949.

Theorem 6.1 (Selberg identity). *For $x > 0$, we have*

$$\psi(x) \log(x) + \sum_{n \leq x} \Lambda(n) \psi\left(\frac{x}{n}\right) = 2x \log x + \mathcal{O}(x).$$

The demonstration of the aforementioned result is detailed, comprising Selberg's portion of the proof in this elementary approach to the PNT. Here, we concentrate on elucidating how this result implies the PNT, which forms Erdős' contribution to the proof. Let $R(x) := \psi(x) - x$ be the *remainder function*. Then we have the following restatement.

Theorem 6.2. *For $x > 0$, we have*

$$R(x) \log(x) + \sum_{n \leq x} \Lambda(n) R\left(\frac{x}{n}\right) = \mathcal{O}(x). \quad (46)$$

We state, without proof, a result of Mertens which we will be used to prove Theorem 6.2.

Fact 6.3 (Mertens, 1874).

$$\sum_{n \leq x} \frac{\Lambda(n)}{n} = \log x + \mathcal{O}(1).$$

Proof of Theorem 6.2. Assuming Theorem 6.1, we have

$$\psi(x) \log x - x \log x + \sum_{n \leq x} \Lambda(n) \psi\left(\frac{x}{n}\right) - x \log x = \mathcal{O}(x).$$

By applying Fact 6.3, we yield

$$(\psi(x) - x) \log x + \sum_{n \leq x} \Lambda(n) \psi\left(\frac{x}{n}\right) - x \left(\sum_{n \leq x} \frac{\Lambda(n)}{n} - \mathcal{O}(1) \right) = \mathcal{O}(x).$$

Hence,

$$(\psi(x) - x) \log x + \sum_{n \leq x} \Lambda(n) \left(\psi\left(\frac{x}{n}\right) - \frac{x}{n} \right) = \mathcal{O}(x).$$

Then using the remainder function, $R(x) = \psi(x) - x$, we have

$$R(x) \log x + \sum_{n \leq x} R\left(\frac{x}{n}\right) \Lambda(n) = \mathcal{O}(x),$$

as desired. □

Theorem 6.2 can be considered as the catalyst for this method to work. The deductions from the Selberg identity to the Prime Number Theorem are credited to Erdős. Observe that (45) is equivalent to

$$\lim_{x \rightarrow \infty} \frac{\psi(x)}{x} = 1 \iff \lim_{x \rightarrow \infty} \frac{\psi(x) - x}{x} = 0 \iff \lim_{x \rightarrow \infty} \frac{R(x)}{x} = 0.$$

However, as per Levinson, proving the above directly from Theorem 6.2 is complicated thus a smoothing function is introduced to replace $R(x)$, this is a similar technique imposed in section (3.4).

Definition 6.4. Define

$$S(y) := \int_2^y \frac{R(t)}{t} dt \text{ for } y \geq 2,$$

and $S(y) = 0$ for $y < 2$.

We now demonstrate the truth of

$$\lim_{y \rightarrow \infty} \frac{S(y)}{y} = 0 \quad (47)$$

suffices to prove the PNT. Let $\varepsilon > 0$ be given. Then by the definition of (47), for sufficiently large y , we have

$$\left| \frac{S(y)}{y} \right| = \frac{|S(y)|}{y} < \frac{1}{3}\varepsilon^2.$$

So it follows,

$$-\frac{1}{3}\varepsilon^2 y < S(y) < \frac{1}{3}\varepsilon^2 y.$$

Hence,

$$\begin{aligned} S(y(1+\varepsilon)) - S(y) &< \frac{1}{3}\varepsilon^2(y(1+\varepsilon)) + \frac{1}{3}\varepsilon^2 y \\ &= \frac{1}{3}\varepsilon^2(y(1+\varepsilon) + y) \\ &= \varepsilon^2 y \left(\frac{2}{3} + \frac{\varepsilon}{3} \right) \\ &< \varepsilon^2 y \quad \text{since } 0 < \varepsilon \ll 1. \end{aligned} \tag{48}$$

Then, by combining the definition of $S(y)$ (as seen in Definition 6.4) with (48), we obtain

$$\begin{aligned} \int_y^{y(1+\varepsilon)} \frac{\psi(t)}{t} dt - \int_y^{y(1+\varepsilon)} 1 dt &= \int_y^{y(1+\varepsilon)} \frac{\psi(t) - t}{t} dt = \int_y^{y(1+\varepsilon)} \frac{R(t)}{t} dt \\ &= \int_2^{y(1+\varepsilon)} \frac{R(t)}{t} dt - \int_2^y \frac{R(t)}{t} dt \\ &= S(y(1+\varepsilon)) - S(y) < \varepsilon^2 y. \end{aligned}$$

Since $\psi(x)$ is, by definition, non-decreasing, we have

$$\frac{\psi(y)}{y(1+\varepsilon)} \int_y^{y(1+\varepsilon)} 1 dt - \int_y^{y(1+\varepsilon)} 1 dt \leq \int_y^{y(1+\varepsilon)} \frac{\psi(t)}{t} dt - \int_y^{y(1+\varepsilon)} 1 dt < \varepsilon^2 y.$$

So it follows

$$y\varepsilon \left(\frac{\psi(y)}{y(1+\varepsilon)} \right) < \varepsilon^2 y + \varepsilon y.$$

Therefore,

$$\frac{\psi(y)}{y} < (1+\varepsilon)^2.$$

Symmetrically, using $S(y) - S(y(1+\varepsilon)) > -\varepsilon^2 y$ from (48), one can deduce that $\frac{\psi(y)}{y} > (1-\varepsilon)^2$ which together implies (45) thus proving the PNT. Observe we did not make any use of Selberg's identity, (46), in the above proof. This suggests it must be used in showing (47) holds.

In fact we have two key features left to complete the Selberg-Erdős elementary proof of the PNT.

- (1) We are yet to prove Selberg's originally proposed identity given in Theorem 6.1. This we will assume true, we refer the reader to Levinson [15, c4] for a complete proof.
- (2) Finally, we must show (47) holds, which we have currently assumed.

6.2 Applying Selberg's identity

The aim of this subsection is to motivate the proof of (47) which states

$$\lim_{y \rightarrow \infty} \frac{S(y)}{y} = 0.$$

We have broken the results given by Levinson [15] into smaller sections for clarity. Proofs that involve tedious or technical manipulations are omitted because our goal is to show how the intermediate lemmas piece together to prove this result. Generally, Levinson simplifies Selberg's identity in multiple stages to construct conditions on a modified version of $S(y)$ which allow us to prove (47). We first prove some properties of $S(y)$, as found in Levinson [15, Lemma 5.1].

Lemma 6.5. *The following are true of $S(y)$.*

(1) $S(y) = \mathcal{O}(y)$, hence there exists $c_1 > 0$ such that $|S(y)| \leq c_1 y$ for $y \geq 2$.

(2) $S(y)$ is Lipschitz, which means $|S(y_2) - S(y_1)| \leq c_2 |y_2 - y_1|$ for some c_2 .

Proof. From Levinson [15, Lemma 3.2], we know $|\psi(x)| \leq \frac{3}{2}x$ for sufficiently large x . Therefore, there exists $M_1 > 0$ such that $|\psi(x)| \leq M_1 x$ for all $x \geq 2$. We subtract x and obtain

$$R(x) = \psi(x) - x \leq M_1 x - x = x(M_1 - 1) \text{ for all } x \geq 2.$$

Hence, dividing through by x gives us $\frac{R(x)}{x} = \mathcal{O}(1)$. Then

$$S(y) = \int_2^y \frac{R(t)}{t} dt \leq \int_2^y \mathcal{O}(1) dt = \mathcal{O}(y), \quad (49)$$

which proves the first result. Although the second part is only used in the simplification of the Selberg identity, we use a similar proof later on hence let us prove this too. Using $\frac{R(x)}{x} = \mathcal{O}(1)$, there exists some $c > 0$ such that

$$|R(x)| \leq cx \quad \text{for all } x \geq 2. \quad (50)$$

Using the first result, $S(y) = \mathcal{O}(y)$, or by applying the fundamental theorem of calculus to (49), we have $S'(y) = \frac{R(y)}{y}$ except at $y = p^i$ where $R(y)$ is discontinuous. Using (50) we can bound the derivative of $S(y)$,

$$|S'(y)| = \left| \frac{R(y)}{y} \right| \leq \frac{cy}{y} = c \quad \text{where } y \neq p^i.$$

Let $y_2 > y_1 > 0$. Suppose the interval $[y_1, y_2]$ contains no prime powers. Then, $S(y)$ is differentiable on this interval and by the mean value theorem, we have

$$\frac{|S(y_2) - S(y_1)|}{y_2 - y_1} = |S(y^*)| \leq c,$$

with $y^* \in [y_1, y_2]$. We can then extend this result to hold for all $y_2 > y_1 > 0$ by applying the triangle inequality around the prime powers where $S(y)$ is not differentiable. This proves our second claim. $\square$

The following proposition, also present in Levinson [15, Lemma 5.1], is presented in our exposition as two separate results. This division underscores the manipulation of the Selberg identity from Theorem 6.2, utilising $R(x)$, into a form employing $S(y)$, our smoothing function.

Proposition 6.6 (First Selberg Manipulation). *We have*

$$S(y) \log y + \sum_{n \leq y} \Lambda(n) S\left(\frac{y}{n}\right) = \mathcal{O}(y). \quad (51)$$

Proof. Recall, from the restated Selberg identity in (46), we have

$$R(x) \log(x) + \sum_{n \leq x} \Lambda(n) R\left(\frac{x}{n}\right) = \mathcal{O}(x).$$

Dividing through by x and integrating both sides from 2 to y gives

$$\int_2^y \frac{R(x)}{x} \log(x) \, dx + \int_2^y \sum_{n \leq x} \Lambda(n) R\left(\frac{x}{n}\right) \cdot \frac{dx}{x} = \mathcal{O}(y). \quad (52)$$

We apply integration by parts to the first expression on the left-hand side of (52) to obtain

$$\int_2^y \frac{R(x)}{x} \log(x) \, dx = \log(y) \cdot \int_2^y \frac{R(x)}{x} \, dx - \int_2^y \left(\int_2^x \frac{R(t)}{t} \, dt \right) \cdot \frac{1}{x} \, dx.$$

Then, using the definition of $S(y)$, the above is equal to

$$\log(y) S(y) - \int_2^y \frac{S(x)}{x} \, dx.$$

Next, we apply Lemma 6.5, which states that $S(x) = \mathcal{O}(x)$, hence $\frac{S(x)}{x} = \mathcal{O}(1)$ to obtain

$$\log(y) S(y) - \mathcal{O}(y). \quad (53)$$

Next, we simplify the second expression on the left-hand side of (52), using a substitution $t = \frac{x}{n}$ as follows

$$\begin{aligned} \int_2^y \sum_{n \leq x} \Lambda(n) R\left(\frac{x}{n}\right) \cdot \frac{dx}{x} &= \sum_{n \leq y} \Lambda(n) \int_2^y R\left(\frac{x}{n}\right) \cdot \frac{dx}{x} \\ &= \sum_{n \leq y} \Lambda(n) \int_2^{\frac{y}{n}} \frac{R(t)}{tn} \cdot n \, dt = \sum_{n \leq y} \Lambda(n) \cdot S\left(\frac{y}{n}\right). \end{aligned} \quad (54)$$

Therefore, by applying both simplifications in (53) and (54) to (52), we have

$$S(y) \log(y) - \mathcal{O}(y) + \sum_{n \leq y} \Lambda(n) \cdot S\left(\frac{y}{n}\right) = \mathcal{O}(y).$$

Hence, by collecting the error terms, we obtain our desired result. $\square$

Levinson then proceeds to apply a series of simplifications to (51) to make the weighted sum more *tractable*. Since this section aims to motivate how Selbergs identity is applied, we refer the reader to Levinson [15, Lemmas 5.2, 5.3, 5.4] for the proofs of the following intermediate steps given below.

Lemma 6.7. *With $\Lambda_2(n) := \Lambda(n) + \sum_{ij=n} \Lambda(i)\Lambda(j)$ and K_1 a constant, we have*

$$(\log^2 y)|S(y)| \leq \sum_{m \leq y} \Lambda_2(m) \left| S\left(\frac{y}{m}\right) \right| + K_1 \cdot y \log y. \quad (55)$$

Lemma 6.8. *There exists a constant K_2 such that*

$$(\log^2 y)|S(y)| \leq 2 \sum_{m \leq y} \left| S\left(\frac{y}{m}\right) \right| \cdot \log m + K_2 y \log y. \quad (56)$$

Lemma 6.9. *There exists a constant K_4 such that*

$$(\log^2 y)|S(y)| \leq 2 \int_2^y \left| S\left(\frac{y}{u}\right) \right| \cdot \log u \, du + K_4 y \log y. \quad (57)$$

In Levinson [15, (4.15)], it was previously demonstrated that $\sum_{n \leq x} (\Lambda_2(n) - 2 \log(n)) = \mathcal{O}(x)$ for $x \geq 2$. This suggests that on average, $\Lambda_2(n)$ behaves like $2 \log(n)$. This observation motivates the simplification from (55) to (56), as $2 \log(n)$ is a simpler expression than $\Lambda_2(n)$. The transition from (56) to (57) involves simply replacing the summation with an integral. This substitution is justified by applying an exponential transformation, as shown in Levinson [15, (5.19)], to the integral form, enabling us to observe the behavior of $S(y)$.

Proposition 6.10. *Let us define the following transformation*

$$W(x) = e^{-x} S(e^x).$$

Then the result in Lemma 6.9 becomes

$$|W(x)| \leq \frac{2}{x^2} \int_0^x (x-v)|W(v)| \, dv + \frac{K_4}{x}.$$

Before proving the above, let us demonstrate the importance of $W(x)$. By showing

$$\limsup_{x \rightarrow \infty} |W(x)| = 0, \quad (58)$$

we deduce $|W(x)| \rightarrow 0$ as $x \rightarrow \infty$ since $|W(x)| \geq 0$. Then, by definition of $W(x)$, it follows that $\frac{S(e^x)}{e^x} \rightarrow 0$ as $x \rightarrow \infty$ which is equivalent to $\frac{S(y)}{y} \rightarrow 0$ as $y \rightarrow \infty$, as stated in (47). This convergence is precisely what we wish to prove. We now prove Proposition 6.10.

Proof. Let $x = \log(y)$ and $v = \log\left(\frac{y}{u}\right)$. Then applying this change of variables to (57) yields

$$x^2 |S(e^x)| \leq 2 \int_0^{x-\log 2} |S(e^v)| (x-v) e^{x-v} \, dv + K_4 x e^x.$$

We can substitute $W(x) = e^{-x}S(e^x)$ to get

$$x^2 \frac{|W(x)|}{e^{-x}} \leq 2 \int_0^{x-\log 2} |W(v)| e^v (x-v) \frac{e^x}{e^v} dv + K_4 x e^x.$$

Dividing through by x^2 yields,

$$|W(x)| e^x \leq \frac{2}{x^2} \int_0^{x-\log 2} |W(v)| (x-v) e^x dv + \frac{K_4 e^x}{x}.$$

Then dividing through by e^x yields,

$$|W(x)| \leq \frac{2}{x^2} \int_0^x |W(v)| (x-v) dv + \frac{K_4}{x},$$

as required. $\square$

The following lemma, as presented in Levinson [15, Lemma 5.5], pertains to the desired quantity discussed in (58), denoted by α . We commence by bounding this with a much more manageable quantity, β .

Lemma 6.11. *Letting*

$$\alpha := \limsup_{x \rightarrow \infty} |W(x)|, \text{ and } \beta = \limsup_{x \rightarrow \infty} \frac{1}{x} \int_0^x |W(t)| dt,$$

we have $\alpha \leq 1$ and $\alpha \leq \beta$.

Observe that (58) implies that all that remains to be shown is $\alpha = 0$. This lemma serves as a stepping stone toward achieving $\alpha = 0$.

Proof. Recall from Lemma 6.5, we have $|\psi(x)| \leq \frac{3}{2}x$. It follows $-x \leq R(x) \leq \frac{x}{2}$. Hence,

$$\frac{R(x)}{x} \leq 1,$$

for sufficiently large x . Therefore, we have

$$\limsup_{x \rightarrow \infty} \frac{|R(x)|}{x} \leq 1. \tag{59}$$

By recalling $S(x) = \int_2^x \frac{R(t)}{t} dt$ for $x \geq 2$ and $S(x) = 0$ otherwise, we have

$$\alpha = \limsup_{x \rightarrow \infty} |W(x)| = \limsup_{y \rightarrow \infty} \frac{|S(y)|}{y} = \limsup_{y \rightarrow \infty} \frac{1}{y} \cdot \left| \int_2^y \frac{R(t)}{t} dt \right| \leq 1,$$

where the final inequality is true by applying (59). To prove $\alpha \leq \beta$, we make use of Proposition 6.10 which was obtained through multiple simplifications in Lemmas 6.7, 6.8 and 6.9. Observe that

$$\begin{aligned} |W(x)| &\leq \frac{2}{x^2} \int_0^x (x-v) |W(v)| dv + \frac{K_4}{x} = \frac{2}{x^2} \int_0^x \int_v^x u \cdot \frac{1}{u} \cdot |W(v)| du dv + \frac{K_4}{x} \\ &= \frac{2}{x^2} \int_0^x \int_0^u u \cdot \frac{1}{u} \cdot |W(v)| dv du + \frac{K_4}{x}, \end{aligned}$$

where the above equality is by reversing the order of integration. Therefore,

$$\frac{2}{x^2} \int_0^x \int_0^u u \cdot \frac{1}{u} \cdot |W(v)| \, dv \, du + \frac{K_4}{x} = \frac{2}{x^2} \int_0^x u \, du \left(\frac{1}{u} \int_0^u |W(v)| \, dv \right) + \frac{K_4}{x}. \quad (60)$$

Observe

$$\frac{2}{x^2} \int_0^x u \, du = 1.$$

Hence, the integral on the right-hand side of (60) can be restated as simply

$$\frac{1}{u} \int_0^u |W(v)| \, dv = \frac{1}{u} \int_0^u \frac{|S(e^v)|}{e^v} \, dv \leq \frac{1}{u} \int_0^u \frac{ce^v}{e^v} \, dv = c, \quad (61)$$

where we apply Lemma 6.5 to bound $|S(e^v)|$. Fix some $x_1 > 0$. Then for $x > x_1$, we can decompose (60) into

$$\begin{aligned} & \frac{2}{x^2} \int_0^{x_1} u \, du \left(\frac{1}{u} \int_0^u |W(v)| \, dv \right) + \frac{2}{x^2} \int_{x_1}^x u \, du \left(\frac{1}{u} \int_0^u |W(v)| \, dv \right) + \frac{K_4}{x} \\ & \leq \frac{2c}{x^2} \int_0^{x_1} u \, du + \frac{2}{x^2} \int_{x_1}^x u \, du \left(\frac{1}{u} \int_0^u |W(v)| \, dv \right) + \frac{K_4}{x}, \end{aligned} \quad (62)$$

where the inequality is obtained by applying (61). Let $\varepsilon > 0$, which we choose later. Then, by the definition of β , for sufficiently large x_1 , we have

$$\frac{1}{u} \int_0^u |W(v)| \, dv < \beta + \varepsilon.$$

Therefore, using this x_1 , (62) asserts that

$$\begin{aligned} |W(x)| & \leq \frac{2c}{x^2} \int_0^{x_1} u \, du + \frac{2}{x^2} \int_{x_1}^x u \, du \cdot (\beta + \varepsilon) + \frac{K_4}{x} \\ & = \frac{cx_1^2}{x^2} + (\beta + \varepsilon) \cdot \left(1 - \frac{x_1^2}{x^2} \right) + \frac{K_4}{x}. \end{aligned}$$

Finally, taking x large, we obtain

$$\alpha = \limsup_{x \rightarrow \infty} |W(x)| \leq \beta + \varepsilon. \quad (63)$$

Suppose, for a contradiction, that $\alpha > \beta$. Then, we can choose ε small enough such that

$$\varepsilon < \alpha - \beta. \quad (64)$$

Then, (64) contradicts (63) so it follows $\alpha \leq \beta$. $\square$

We need two more facts that will enable us to prove $\alpha = 0$ and thus conclude the PNT. These results correspond to Levinson [15, Lemmas 5.6, 5.7].

Lemma 6.12 (Lipschitz Property). *Let $k = 2c$. Then $W(x)$ is Lipschitz with Lipschitz constant k , so for $x_1 < x_2$ we have*

$$|W(x_2) - W(x_1)| \leq k|x_2 - x_1|.$$

Proof. Using the definition $W(x) = e^{-x}S(e^x)$, we can apply the product rule and obtain

$$|W'(x)| \leq |-e^{-x}S(e^x) + S'(e^x)| \leq e^{-x}|S(e^x)| + |S'(e^x)| \leq e^{-x} \cdot e^x c + c = 2c = k \quad \text{where } e^x \neq p^i.$$

We first apply the triangle inequality and then use Lemma 6.5. Then, by a similar argument as seen in Lemma 6.5, we conclude the result. Bounding the derivative is essentially sufficient; the rest of the argument follows from the mean value theorem and triangle inequality. $\square$

Lemma 6.13 (Bounding Lemma). *Let $v_1 < v_2$. Suppose $W(v) \neq 0$ for $v \in (v_1, v_2)$, then there exists $M > 0$ such that*

$$\int_{v_1}^{v_2} |W(v)| \, dv \leq M.$$

Proof sketch. The proof involves an application of the Abel summation formula (see Appendix (B)) and an intermediate result in Mertens's second theorem, specifically Fact 6.3. $\square$

The aforementioned two facts, along with Lemma 6.11, are used to establish $\alpha = 0$, thereby proving the PNT. We conclude with our essay with precisely this result.

Lemma 6.14. *A function $W(x)$ subject to Lemmas 6.11, 6.12, 6.13 must have $\alpha = 0$.*

Proof sketch. Firstly, choose $\gamma > \alpha$. Then by the definition of α , there exists an x_γ such that

$$|W(x)| \leq \gamma \quad \text{for all } x \geq x_\gamma. \quad (65)$$

The proof, detailed fully in Levinson [15, (5.8)], divides into two cases concerning the distribution of zeros of $|W(x)|$.

Case 1: Suppose $W(x) \neq 0$ for all $x \geq x_\gamma$, it then follows from Lemma 6.13 that

$$\int_{x_\gamma}^x |W(v)| \, dv \leq M, \quad (66)$$

for some $M > 0$ and for all $x \geq x_\gamma$. Therefore, we have

$$\frac{1}{x} \int_0^x |W(t)| \, dt = \frac{1}{x} \left(\int_0^{x_\gamma} |W(t)| \, dt + \int_{x_\gamma}^x |W(t)| \, dt \right) \leq \frac{1}{x} \left(\int_0^{x_\gamma} |W(t)| \, dt + M \right),$$

where the last inequality uses (66). The expression above tends to 0 as $x \rightarrow \infty$ which implies $\beta = 0$ in Lemma 6.11. Using $\alpha \leq \beta$ we conclude $\alpha = 0$ in this case.

Case 2: Otherwise, let $W(x)$ have arbitrarily large zeros. Levinson proceeds to consider intervals between successive zeros and bounds them individually using the bounding lemma 6.13, alongside the Lipschitz property of $W(x)$ in Lemma 6.12. This process is carried out in three further sub-cases, the details of which can be found in the original proof. After careful consideration of each sub-case, we conclude that all sub-cases imply

$$\frac{1}{y} \int_0^y |W(t)| \, dt \leq \frac{1}{y} \int_0^{x_1} |W(t)| \, dt + \gamma \left(1 - \frac{\alpha^2}{2Mk} \right) + \frac{M}{y}.$$

Now, similar to *Case 1*, letting $y \rightarrow \infty$ implies $\beta \leq \gamma \left(1 - \frac{\alpha^2}{2Mk}\right)$, and using $\beta \geq \alpha$ from Lemma 6.11, it follows

$$\alpha \leq \beta \leq \gamma \left(1 - \frac{\alpha^2}{2Mk}\right).$$

Finally, since this holds for all $\gamma > \alpha$, it must hold for $\gamma = \alpha$ and thus by a simple substitution and rearranging, we obtain $\alpha^3 \leq 0$. Using $\alpha \geq 0$ from the definition of α , we conclude $\alpha = 0$ which concludes this elementary proof of the PNT. $\square$

7 Conclusion

In this essay, we have presented two distinct proofs of the Prime Number Theorem. The first employs the traditional complex analytic approach, focusing on the zeros of the Riemann zeta function, while the second adopts a more elementary method that does not rely on the complex analytic properties of the zeta function. Similarly, our exploration of the gaps between consecutive primes mirrors this duality: we discussed results by Hoheisel, Chudakov, Montgomery, and Huxley, which rely on zero density estimates of the Riemann zeta function, alongside the recent breakthrough by Baker, Harman, and Pintz, which leverages modern sieve theory to achieve slightly stronger results. We have also delved into well-known conjectures such as Lindelöf's and Cramer's, demonstrating how, under these strong conditions, we can derive much stronger implications regarding gaps in consecutive primes.

Appendices

A $\zeta(z)$ holomorphic except at $z = 1$

We first let $U = \{z \in \mathbb{C} : \Re(z) > 1\}$ and consider $\zeta_0(z) : U \rightarrow \mathbb{C}$ given by:

$$\zeta_0(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}$$

Then we have for $z = \sigma + it$ with $\sigma > 1$

$$\sum_{n=1}^{\infty} \left| \frac{1}{n^z} \right| = \sum_{n=1}^{\infty} \frac{1}{n^{\Re(z)}} \leq \sum_{n=1}^{\infty} \frac{1}{n^{\sigma}} < \infty$$

and the final sum converges due to the p-series test. Since this is local uniform convergence of a sum of holomorphic functions on U , we have ζ_0 defined above is holomorphic on U . We now extend ζ_0 using the following theorem from Ahlfors [1]

Theorem A.1. *There exists a holomorphic function $\zeta : \mathbb{C} \setminus \{1\} \rightarrow \mathbb{C}$ such that $\zeta|_U = \zeta_0$ and $z = 1$ is a simple pole of ζ with residue 1.*

This shows the Riemann zeta function is holomorphic on $\mathbb{C}$ except at $z = 1$ where it has a simple pole with residue 1.

B Abel Summation Formula

The Abel Summation formula, attributed to Niels Henrik Abel, is a tool used in analytic number theory to compute series and study special functions. We give the formulation used in Levinson [15, (3.1)].

Theorem B.1. *Let $f(t)$ have a continuous derivative, $f'(t)$, for $t \geq 1$. Let c_n , $n \geq 1$, be constants and let $C(u) = \sum_{n \leq u} c_n$. Then*

$$\sum_{n \leq x} c_n f(n) = f(x)C(x) - \int_1^x f'(t)C(t) dt$$

and

$$\sum_{n \leq x} f(n) = \int_1^x f(t) dt + \int_1^x (t - [t]) \cdot f'(t) dt + f(1) - (x - [x]) \cdot f(x)$$

The result is proved by applying integration by parts to a Riemann-Stieltjes integral with the functions C and f .

C Datatable concerning $p_{n+1} - p_n < p_n^k$

The following table provides an explicit value of N such that for $n > N$, $p_{n+1} - p_n < p_n^k$ after checking the first 1,000,000,000 prime numbers.

k	N	P_N, P_{N+1}
0.7	5	11, 13
$\vdots$	$\vdots$	$\vdots$
0.6	5	11, 13
$\vdots$	$\vdots$	$\vdots$
0.58	5	11, 13
0.57	10	29, 31
0.56	10	29, 31
0.55	31	127, 131
$\vdots$	$\vdots$	$\vdots$
0.5	30	127, 131
0.49	218	1361, 1367
$\vdots$	$\vdots$	$\vdots$
0.43	218	1361, 1367
0.42	264	1693, 1697
0.41	3386	31469, 31477
0.4	3385	31469, 31477
0.39	3645	34123, 34127
0.38	4523	43391, 43397
0.37	14358	156007, 156011

k	N	P_N, P_{N+1}
0.36	40934	492227, 492251
0.35	40934	492227, 492251
0.34	149690	2010881, 2010887
0.33	149690	2010881, 2010887
0.32	325853	4652507, 4652513
0.31	1319946	20831533, 20831557
0.3	2850175	47326913, 47326919
0.29	6957877	122164969, 122164981
0.28	23163299	436273291, 436273309
0.27	36681600	708730541, 708730577
0.26	203615628	4302407713, 4302407729

Table 1: Data computed using Python and SageMath concerning various values of k .

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